

Article

A Framework for Improving Accessibility of Serious Games in Handheld Augmented Reality Based on User Interaction Data

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Abstract: Serious games combined with augmented reality (AR) can positively impact user's motivation and learning experience. To ensure that users with different needs can use them effectively, we must ensure the accessibility of such solutions. The goal of this paper is to show that game-learning analytics can be utilized to identify issues for people with various needs and define ways to improve the accessibility of AR-based serious games. This paper presents a framework based on analysing automatically collected user interaction data in handheld AR-based serious games to adapt them to the user in the context of accessibility. An experiment was conducted with people with disabilities and people of different ages using the established framework. The focus of the experiment results analysis in this paper is on the evaluation of interaction mechanisms adapted to users based on their interaction data and preferences. The results showed that for most participants the assigned interaction mechanisms for virtual object transformation tasks (translation, scaling, rotation) in AR were satisfactory. The framework proposed in this paper provides the basis for further research in this field concerning the use of advanced analytics that enable intelligent adaptation to the individual user.

Keywords: serious games; augmented reality; accessibility; universal design; user interaction; adaptation to the user; game-learning analytics



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1. Introduction

Augmented reality (AR) is an emerging technology that supplements reality and allows the user to experience the real world with virtual objects superimposed on or composed with the real world [1]. One of the key advantages of AR is that it is available on most of today's smartphones and tablet devices, which makes it more available to a wider population of users [2]. Mobile AR has already proven its potential in various application areas, such as retail/shopping, interior design, tourism, navigation, training, education, healthcare, military, video games, and others [3]. As an emerging technology, it enables new forms of human–computer interaction compared to traditional mobile solutions and provides users with engaging experiences, which has a positive impact on users' motivation to use AR-based solutions. This is especially recognized in the field of game-based learning, as learners nowadays require new and more interesting learning experiences [4]. The concept of games that are not played primarily for amusement, but to impart specific knowledge or skills to the player, is known as “serious games” and was first coined by Abt in 1970 [5]. According to [6], the term “serious games” should be used for games explicitly designed for purposes other than entertainment, i.e., they can be seen as having a specific purpose and not as a synonym for other existing applications. Over the years, serious

games have experienced growth, especially in domains of education, training, well-being, advertisement, cultural heritage, interpersonal communication, and healthcare [7].

Serious games can be considered effective educational tools when applied in education properly and in a student-centred way, which is considered one of the key features of the new generation of students [4]. AR-based serious games have also proven their potential to increase student interest and influence motivation and learning efficiency [8,9]. Great interest has also been shown in the field of inclusive education, where research can be found focusing on the design of serious games for a wide range of students with disabilities. However, the development of such solutions is still more focused on a specific difficulty or disability, without applying the principles of Design for All [10].

To benefit all learners, serious games must be developed to meet the needs of a wide range of users, including users with disabilities. This is especially important for AR-based serious games which involve different interactive tasks with virtual content in the real environment, such as those identified in [11]. Intuitive and user-friendly interfaces are also critical for the successful implementation of AR in educational contexts [12]. Thus, AR-based solutions are extremely suited for exploring the best ways of interacting with the user depending on their needs, limitations and context, i.e., to make them more accessible as there are still many accessibility challenges that need to be addressed [13,14]. Accessibility is a measure of the extent to which a product, service, or environment is suitable for all users, including people with disabilities and the elderly. When considering users with disabilities, accessibility can be considered a great part of usability, as it is defined as the “extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use” [15]. It can generally be achieved by applying the principles of Universal Design (UD), which is a design method that aims to integrate the needs of people with disabilities and the elderly into products intended for widespread use. Besides addressing accessibility needs, UD addresses the needs of all people to the greatest extent possible, without the need for adaptation or specialized design [16], including people with different ages, sizes, abilities or disabilities. One of the UD principles is the *flexibility in use* principle that aims to provide users with choices, allowing them to interact with and adapt products or environments according to their unique needs, pace, precision, and control [17].

Nowadays, it is already widely accepted that serious games, integrated as an aid to learning different subjects, play an important role in learning and help students focus on the learning goal [18]. Some of the factors influencing the effect of serious game-assisted learning are the perceived usefulness, ease of use, clarity of the goal, as well as adaptivity to the needs of the user [18]. Additionally, it was shown that serious games with a higher mental load have a negative impact on the learning effect [19]. Since AR enables new ways of interacting, AR-based serious games must be designed in a way that is intuitive and not mentally demanding. In this context, they should not require high physical effort or cause frustration and should be adaptable to the needs of individual users. Because there are users who cannot or do not know how to adapt the content and aspects of the user interface (UI) to meet their needs themselves, it is essential to enable ways of adapting the game without their intervention, i.e., to automatically adapt the game based on their needs. Examples of users who might benefit from automatic adaptation are users who, due to various disabilities, cannot adapt the solution themselves (e.g., people with severe motor disabilities), older people who are not so familiar with new technologies, children who often do not understand how certain settings can be useful for them, as well as users with no prior experience with AR in general. Due to heterogeneous needs, as well as the often-conflicting requirements of said users, the field of automatic adaptation in the context

of accessibility is still an under-researched area [20]. Although there has been some success, it has been difficult to achieve robust automatic personalization [20].

Researchers are encouraged to continue the approach of qualitative and quantitative evaluation to assess the usability of AR applications, as well as to utilize different methods to better understand how a user interacts with the system, such as by capturing as many different performance metrics from the user as possible (i.e., not just task-completion time) [21]. When defining accessible interaction techniques and content display for a certain user, different factors must be considered. For example, which input method is suitable to perform a certain task in the AR solution, and how various contextual settings are impacting the accessibility, such as environmental factors and type of the device used [22]. Given the increasing popularity of serious games and game-based learning, game-learning analytics provide one way of tracking different parameters to analyse user behaviour and evaluate performance which enables suggestions on the content and interaction customization to facilitate game use to the individual user.

In this paper, we propose a solution for enhancing the accessibility of handheld AR-based serious games by leveraging existing game-learning analytics platforms, given the benefits that serious games offer to learners. Tracking user interaction data and various contextual parameters enables a deeper understanding of how users interact with AR systems. By adapting content and interaction mechanisms to fit users' needs, we move closer to solutions designed around the principles of universal design, fostering social inclusion for diverse users, including children, people with disabilities, and the elderly.

The main contribution of this research is a framework that enables the automatic tracking of user interactions in AR-based serious games for mobile devices in the context of accessibility, facilitating data analysis and adaptive accessibility settings tailored to individual users. To the best of our knowledge, no existing framework integrates AR with adaptive accessibility while utilizing methods and tools from game-learning analytics. Our framework addresses this gap by enabling the automated tracking of user interaction data, mapping user abilities and/or preferences to interaction parameters, and dynamically adjusting accessibility settings to improve usability for individuals with various disabilities and of different ages.

Furthermore, while accessibility in AR-based games is an emerging field, existing solutions often lack real-time adaptability and personalized accessibility adjustments. Our approach fills this gap by introducing a methodology for automated data collection, storage, and analysis of interaction parameters within the accessibility context, allowing for customized content and interaction mechanisms based on individual needs. Additionally, the flexibility of our framework allows its application beyond AR-based serious games, extending to digital games in general and contributing to broader digital inclusion efforts.

The sections of this paper are organized as follows. Section 2 introduces the topic of user needs in AR in the context of accessibility as well as investigates the current state in the field of adaptation to the user in AR and tracking user interactions in serious games. In Section 3, we propose a model for tracking user interactions in AR-based serious games in the context of accessibility. Section 4 then presents the methodology for automated collection, storage and analysis of user interactions. The implemented technology platform that is based on the proposed model and methodology is then described along with the proposed framework and its key parts. An experiment with user testing is described in Section 5 along with a discussion of the results. Conclusions and future research are presented in Section 6.

2. Related Work

In this section, first, we give an overview of user needs in AR. Next, we give an overview of studies related to the field of adaptation of AR to the user in the context of accessibility. Afterwards, we present relevant work dealing with serious games and game-learning analytics.

2.1. User Needs in Augmented Reality

A key approach to achieving accessibility is user-centred design, which is based on active user involvement to improve understanding of user goals and needs, as well as iterative development and evaluation of solutions [23]. Some of the ways to understand user needs in immersive environments in terms of accessibility are to conduct usability tests with an experienced user test facilitator or build platforms with active input from users with diverse needs [24]. These methods and interactions with real users can provide designers and developers with insight that will guide their design and development decisions [24].

Although there are various initiatives for the accessibility of virtual reality (VR) and AR technologies, commercial systems using these technologies are still not widely adopted by people with disabilities and the elderly [25]. One of the main reasons is the lack of standardized accessibility guidelines for developers and designers. A working group of the World Wide Web Consortium published a draft document called XR Accessibility User Requirements (XAUR) [26] with a description of the needs and requirements of people with disabilities as users of VR, AR and other immersive technologies. Multimodal support and adaptation of content and user interface to different user needs are highlighted as important characteristics of accessibility [26]. The literature review related to XR technology performed in [27], found that one of the general challenges of XR technology is lack of multimodality and compatibility with assistive technologies as well as customization options that could benefit their specific special needs [27]. The challenges emphasized in [27] suggest that the universal design of XR technology has not been sufficiently discussed and that there is an urgent need to make XR technology more accessible and user-friendly to all people.

Along with the understanding of which input and output modality best suits the user, it is important to distinguish all the other less obvious factors that affect the usability and user experience of solutions based on immersive technologies. For handheld AR solutions, some of the factors may be, for example, sound clarity, text size, colour contrast, key layout options, environmental movement requirements, interaction with the environment or the merging of virtual objects with the real world. Although touch gestures on touchscreens are one of the widely used interaction methods in handheld AR, this method can often lead to imprecise interactions or unexpected errors due to inaccurate hand coordination [28]. Therefore, additional techniques have been investigated to improve the precision of interactions, e.g., a freeze technique that allows the user to freeze an AR display or a virtual object to interact with it [28]. These and many other factors influence whether some AR solution is an accessible and enjoyable experience for users regardless of their abilities and/or limitations. Once user needs are recognized in the context of accessibility features, it is important to understand how to implement and incorporate them into the solution or, in the case of accessibility options, where to place them so that they are always available (but not obtrusive on the screen) [26].

One of the findings from the study in [29] is that, alongside the guidelines, most XR developers would prefer platform-targeted, open-source packages they can drag and drop into their projects or pull specific features from. An example of an accessibility framework designed for developers of XR applications is described in [30], and while it offers different customizable features, mostly for blind and low-visioned users [30], it is not yet made

publicly available to developers. Providing accessibility options to change the appearance of the user interface (UI) and AR in terms of colour (contrast), font type, and size, would be beneficial for all age groups, including children and older users. For younger users, it is also important to provide multiple means of representation, especially for solutions intended for learning, while for adults it is important to ensure support for assistive technologies or to provide alternative input methods [22]. Personalization at the individual level is a very complex task, as the requirements of different users can be contradictory, even if they belong to the same age or disability group (e.g., a group of users with intellectual disabilities [31] or with dyslexia [32]).

2.2. *Adaptation to the User in Augmented Reality*

The importance of adaptation to user needs in AR-based solutions is recognized by many researchers. A case study described in [33] adapts the AR-based user interface for older users to facilitate their everyday work on complex automated machines. The UI components are adapted based on a set of knowledge-based configuration rules that consider their skills, tasks, age, and cognitive and physical abilities. Adapting to the user's needs in a given context in real time has also been important for security reasons as well, as described in [34] where an AR-based application helps navigate people with mild intellectual disabilities when they go out of the permitted area.

Although research on specific AR applications for people with disabilities is useful, more research effort needs to be put in terms of adaptation of AR interaction mechanisms and content so that they can be used to the greatest extent possible by all people regardless of their age or disability. Research related to the adaptation of interaction and content in terms of accessibility has mainly been conducted for traditional audio-visual media (not immersive media) [25]. Furthermore, interface personalization has mainly been based on the user profile and explicitly stated preferences, without objective mapping between users' range of abilities and interface parameters, especially not in solutions based on XR technologies [25]. This is also emphasized by participants from a study in [14] where they recommended tailoring immersive experiences around the abilities of the user and considering the "powers" of people with impairments (such as heightened senses, superior multi-tasking skills, and higher attention to visual details). One of their suggestions is to develop an immersive environment that dynamically maps the reality/disability of the user and simplifies movements by mapping the physical abilities of users and certain movements to virtual spaces [14].

The concept of personalization is closely related to the concept of adaptation. According to [35], the (automatic) adaptation is the key to accessibility for the ever-increasing diversity of users in the human-computer interaction field, where adaptation is defined in terms of how the software can accommodate the specific physical and cognitive abilities of the user, as well as the situation of use and platform capabilities [36]. Web content developers have long been aware of the need for personalized content for the user, and many Internet portals already offer simple tools for filtering information. However, most of these tools rely on manually adapting these options, which is becoming an increasingly tiresome process for users. There is an obvious need for increased personalization in many areas of interactive software, especially in ways to automate this process, which suggests the use of machine learning techniques to personalize interfaces [37]. An example of a framework that proposes both rule-based and machine-learning approaches to make AR interface design relevant to specific contexts is described in [38]. However, it should be investigated how it could be utilised to develop more adaptive interfaces in the context of accessibility. As pointed out in [38], extensive data collection for training is required

for implementing intelligent AR, which is a significant practical challenge that needs to be addressed in the future.

Although the implementation of different questionnaires (e.g., game user experience evaluation or standardized usability test questionnaires) can provide general insight into the user experience of the game and the perceived usability of the solution, to identify more information about technical, gameplay and usability issues, it is beneficial to include qualitative methods as well, e.g., interviews [39]. In addition to conducting interviews with users to more accurately identify the elements that affect playability, i.e., the pleasantness and ease of use of the game, monitoring different objective measures in the game could help identify more specific usability problems, as well as different user preferences (if subjective evaluation is integrated into a game) [22]. Objective metrics that are usually analysed in research related to the usability of AR solutions and when comparing different interaction techniques in AR are: performance time, number of tracking losses, time to recover tracking, time to select, number of errors or misalignments, number of requested instructions in the game, number of successful tasks or number of failures [40–43]. On the other hand, for the concept of adaptation of AR content, researchers in [44] considered parameters related to the environment, i.e., the scene illumination, the distance to the target and the ambient noise, but the importance of adapting AR with respect to the user and the platform is also emphasized.

Some research studies based the adaptation of the AR content on user behaviour in different contexts. For example, in [45], based on the past user behaviour and feature data extracted from the built-in sensors of the smart glasses, the researchers adapted the augmented content on smart glasses (e.g., specific point-of-interest information while moving). In [46], the researchers proposed an AR traffic model whose aim is to improve the quality of experience of AR services depending on the user's movement. However, there is a lack of studies related to the automatic adaptation of AR solutions to the user in terms of accessibility. The interaction model for AR systems proposed in [47] takes into account the different abilities of users, but the component related to the UI adaptation is based on preferences explicitly given by the user, i.e., individual parameters such as cultural preferences and colour schemes, are manually adjusted. To get closer to dynamically determining accessibility features that address the user's needs, it is necessary to find a way to objectively map the user capabilities and UI parameters, as mentioned in [25].

2.3. *Serious Games and Game-Learning Analytics*

To encourage the use of serious games as a learning tool, there is a need to develop a deeper understanding of how they affect the learning process, what skills and techniques they can provide, and how to align them with learner preferences [48]. Knowledge of the above can be gained by evaluating the effectiveness of games. An increasingly popular approach to evaluating the games is using the games themselves as an evaluation tool. The collection of basic information has always been carried out using Learning Management Systems, which provide a certain degree of insight into learner activities. Detailed gameplay tracking can contain valuable information about how users played the game, and where they encountered different obstacles or problems understanding key concepts, which can be useful for teachers but can also provide valuable feedback in the context of the game design itself. Thus, by in-depth analysis of the interactions of users of serious games, it is possible to provide a better understanding of how users use serious games and how to make them more effective [48].

Analysing user behaviour through big data analytics is a trend that has seen very rapid growth in recent years, especially in the social media domain. The trend was also reflected in the domains of education and video games affecting each domain to develop

its techniques and tools for analysing user interactions [48]. Learning analytics tools can bring numerous benefits in the field of education in the context of improving student retention in schools, increasing student success, and creating personalized learning [49]. Artificial intelligence techniques for learning analytics are gradually being adopted in the field of education to improve the teaching and learning process, and to support teaching activities at different educational levels [50]. Although serious games for education have been shown to have an effective impact on students' behaviour, learning, and emotions, further research related to data science techniques is needed to improve the usability of such games [51]. The use of AR applications in education is gaining interest at all levels of education and in all contexts. However, efforts to integrate learning analytics techniques into immersive technologies are limited and rare [52]. Nevertheless, researchers are encouraged to target their research to investigate more how learning analytics impacts immersive technologies [53]. Additionally, the integration of AR and learning analytics raises new questions related to privacy and ethics, which is particularly important for users with different disabilities who need to be provided with additional support and guidance [54].

While the goal of learning analytics is to improve learning performance in various online environments, game analytics is used to improve the user experience of the game. Game analytics is a term used in the video game industry for the application of analytics in game development and research to better understand how users play games, find issues, and improve the gaming experience [55]. Most game analytics systems monitor user data, which can be data in the context of the user as a buyer of a game/component in the game or a member of various game-related communities, or game metrics, i.e., everything related to the direct interaction of the user with the game [48].

For serious games that are used for educational purposes, it is necessary to monitor both data related to the learning process and data related to the game. The goal of analysing this data is to understand how the user learns and how the game should be designed to achieve the best possible result when a user interacts with it. This can be achieved by combining methods of learning and game analytics, i.e., using the game-learning analytics approach [48].

According to findings from a review in [56], learning analytics has the potential to support adaptive learning in game-based learning by employing artificial intelligence and educational data mining techniques. However, besides adapting the learning process based on learners' preferences and interests, the learners' needs in terms of accessibility must be considered as they have a direct impact on game effectiveness [56]. One of the main benefits of an accessible video game is the inclusion of all types of users. However, ensuring accessibility also affects improved access to learning content and, consequently, achieving better learning outcomes. It also allows users with temporary or permanent disability to receive and understand educational content as well as the reuse of content on multiple devices [57].

To summarize, existing game-learning analytics platforms allow different metrics and user data to be collected and used for different analyses and purposes. In this paper, game-learning analytics methods are combined with the best approaches from the literature for evaluating the usability of serious games in AR to identify accessibility issues for people with various needs.

3. Model for Tracking User Interactions in AR-Based Serious Games

In this section, we first describe the conceptual model for tracking user interaction data in AR-based serious games with features that are important to track in the context of accessibility. In the context of adaptation to the user, features here refer to user char-

acteristics as well as measurable interaction parameters that provide insights into user engagement, behaviour, and accessibility-related challenges in AR-based serious games. Afterwards, we specify the data model for tracking user interactions in more detail to enable its usage in existing game-learning analytics systems to get insight into usability problems and enhance accessibility.

3.1. Conceptual Model

A model for tracking user interactions in the context of accessibility of AR-based serious games for mobile devices proposed in this section defines how user interaction data are collected and stored in a structured way. User interaction data in this case refer to all user activities, behaviour and user preferences collected during interaction with an AR serious game. The model is based on event-based tracking as one of the most widely used methods to track user interactions in serious games [58]. This model aims to enable easier data analysis for understanding the data to draw relevant conclusions, such as how to improve and adapt an AR-based serious game in the context of accessibility for users.

The approach to tracking user interactions in this context is not unambiguously determined. Various factors can affect the accessibility of such solutions, and they affect what data need to be tracked during user interaction. For this reason, we first describe the features on which the user interaction tracking model is based, and then the model is specified in more detail. The conceptual data model for tracking user interactions is shown in Figure 1. Tracking user interaction, here represented as an event, is based on the features of the next five components: User, Serious Game, Augmented Reality, Context, and Accessibility. The importance of each component and its features is described below.

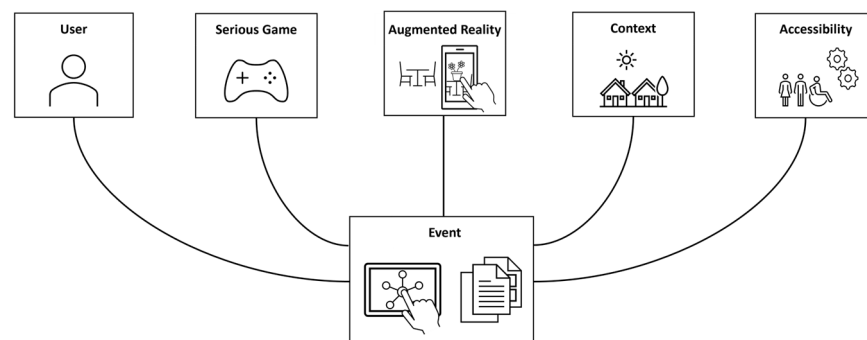


Figure 1. A conceptual model for tracking user interactions based on features related to the user, serious game, augmented reality, context and accessibility.

3.1.1. User

To enable objective mapping of user characteristics and interaction parameters to adapt the solution to the individual in the context of accessibility, one must develop a user model that represents preferences, capacities, and affective processes and their relationship with the task at hand [59]. In other words, depending on the context in which the solution is to be adapted to the user, different characteristics of the user are considered. In the context of accessibility, it is extremely important to consider the disabilities that the user has for defining the user interaction tracking model.

Having said that, the features that one user profile could contain to help determine the direction of the adaptation of the interaction mechanisms and AR content are, for example, age, gender, type of disabilities, previous experience with technology, and user preferences. Such a user profile can be created during the registration process and can be dynamically upgraded depending on the user interaction data. If subjective evaluation is integrated into

a game, tracking subjective metrics could help identify user preferences related to different game elements.

The following are examples of how a particular difficulty or impairment a user has can affect the adaptation of a solution to improve accessibility:

- physical disabilities—understanding the user’s physical disabilities can be crucial for adapting the interaction mechanisms and interface of AR-based solutions. This includes assessment of fine motor skills, hand mobility, coordination and the ability to follow movements. This information can help adapt ways of interaction by suggesting input methods that best suit a user’s need, such as gestures or voice commands;
- visual impairments—users may have different visual impairments. Therefore, factors such as visual acuity, colours visible to the user, sensitivity to light or contrast, and other visual impairments, should be considered. This information can help to adapt the AR virtual objects, but also UI elements to make them more visible to the user;
- auditory impairments—there are users with varying degrees of auditory impairment. Adapting to such users may therefore involve providing alternative means of communication such as text displays or vibrations, or adapting the sound and sound elements to the user’s needs;
- cognitive, learning and neurological disabilities—each user also has different cognitive abilities and understanding. To improve accessibility, comprehension, concentration, memory and other cognitive factors must be taken into account. The user interface and interaction should be simple, intuitive and customised so that users can easily use the solution.

3.1.2. Augmented Reality

Since the focus in this work is on serious games for handheld devices, the AR features to consider for the user interaction tracking model are related to the tasks and interaction techniques commonly appearing in such solutions. These tasks can be grouped into the following five categories: observing AR content, establishing physical/virtual correspondence, creating virtual content, transforming virtual content, and activating virtual content [11]. Some of the interaction techniques for AR tasks identified in an overview of AR solutions developed for people with disabilities [22] are as follows: pointing the device’s camera to a marker/image/text, directing the device’s camera to the point of interest/location, tapping to select or position the object, drag and drop to translate the object, swipe to rotate, pinch to zoom to resize, click to change the colour, double-tap on an object to activate it, etc. One of the first steps in most AR solutions is to establish an AR scene by scanning the environment or a marker/image on which the virtual object appears. This would be considered a crucial event to be tracked to identify if a user has issues. By tracking parameters related to this task, such as the number of scanning attempts, the duration of the task and the size of the recognized surface, one can get an insight into the usability of this task in general, or accessibility measures needed for an individual user. Similarly, it is possible to define interaction parameters related to other tasks in AR.

Table 1 brings examples of parameters to track in the events related to a particular AR task. The parameters listed in the table are the most often tracked objective metrics in software solutions to detect usability problems, especially in research studies that compare different interaction techniques for performing AR tasks [40–43,60]. In addition, these parameters are most often tracked in tasks related to 3D manipulation of objects, which represent one of the most critical tasks of mobile AR.

Table 1. Example metrics (parameters) that can be tracked in a particular AR event (task).

AR Event	AR Task Type	Tracking Parameters
Observing AR content	Object observation	Observation time Time outside the camera's field of view Distance from the object
Establishing physical/virtual correspondence	Marker/location/object/surface scanning	Time for scanning Number of attempts Surface size Distance to point of interest
Creating virtual content	Object positioning	Time for positioning Position on the device screen
Transforming virtual content	Translating Rotation Scaling Colouring	Transformation time Object size Interaction type Number of each interaction Distance from the object
Activating virtual content	Object selection Activating the effect	Activation time Number of attempts Interaction type

In AR for mobile devices, the most common input method for interaction with virtual objects is touch gestures, i.e., certain finger movements on the touch screen. The type of touch gesture used depends on the task. For example, a touch gesture is used to select or place a virtual object in the environment, a drag and drop gesture is used to translate a virtual object, a pinch gesture is used to scale a virtual object and rotate two fingers (counter) clockwise gesture is used to rotate a virtual object. Certain touch gestures can be implemented in many ways, e.g., to rotate an object, in addition to the abovementioned gesture, one can also use one finger in the direction in which you want to rotate the object or rotate the object using two thumbs (using two hands) [61]. Although most applications use touch gestures, in some cases it is suggested to use indirect controls, i.e., buttons on the screen, in such a way that the user does not need to change the way of holding the device, and it is recommended to use the transparency of the buttons to avoid obscuring the AR scene [62]. Although the use of transparent buttons is recommended, it should be considered that for some users they may not be sufficiently visible, so the display of the UI should be adapted accordingly. In case when there is more than one interaction technique implemented for the same task, those techniques must not be conflicting [60]. Considering the above, in the model for tracking user interaction it is necessary to distinguish the type and number of interactions that the user used in the given AR task.

Depending on the type of serious game, the interaction parameters should be appropriately identified. For example, in addition to the mentioned AR tasks (select or position, translate, rotate and scale), some games will require tracking an event of *activating virtual content*. It can be triggered by, e.g., a swipe-up gesture for a jumping effect. Also, for a more accessible approach, one should ensure an alternative mechanism of interaction, which in the example of the jumping effect can be clicking the up button.

3.1.3. Serious Game

Since the goal of the serious game is to improve existing or acquire new knowledge and skills of the user, tracking events that represent progress in the game and timestamps when they occurred can indicate the user's progress in a serious game, as well as in learning. Also, tracking in-game interactions can help detect user behaviour and reveal potential

usability and accessibility issues (e.g., exiting the game too often, skipping a certain task or asking for help).

The interaction model proposed in [58] derived from the analysis of how learners' interactions are tracked in serious games. In this interaction model, events are used to represent players' interactions in a game where an interaction event is defined with the following attributes: (1) a timestamp, representing the instant the event was generated in the game; (2) a user ID, identifying the player that generated the event; (3) an action, representing the type of interaction performed by the player; (4) a target, representing a game element that is the objective of the player's action; and (5) an optional value, representing the parameters of the action [58]. The interaction model defined like this suits the needs of tracking user interactions in AR-based serious games in the context of accessibility as well. Table 2 shows common interactions tracked in a serious game represented by a target (objective of player's action) and an action (player's interaction type), while metric examples are listed as one that can be tracked in AR-based serious games also. The model can be extended with custom interactions that can be generalized and shared across many serious games (e.g., AR-based serious games), along with the automatic processes to analyse them [58] (e.g., to enhance accessibility if issues are detected).

Table 2. Common events tracked in a serious game represented by a target and an action with metric examples (adapted from [58]).

Target	Action	Metric Example
Completable (game, game session, level)	Start Progress Complete	Level of completion Time to complete Number of plays Finish time
Alternatives (in-game decisions related to questions, menus, paths)	Select	Selected options Time to choose/answer Ratio of correct/incorrect answers
Meaningful variables (result, position, jump, game condition)	Set	Final value Value's evolution
Custom interactions	(game dependent)	Number of requests Request type (e.g., help)

To take advantage of game-learning analytics (GLA), all components of the GLA system must be implemented so that they support the process from data collection to the adaptation of the game to the individual user. A successful GLA system requires integration with existing learning analytics platforms and game engines. Since the domains of learning and gaming analytics have been developed separately and with different goals, the communication methods between the components of such systems differ, and the problem of interoperability appears [48]. So, it is essential to use a standardized format for data collection to ensure interoperability of platforms. As shown in [58], the proposed interaction model can be implemented using the Experience API (xAPI) standard which is designed to track user's activity in the form of sequential streams [63].

The xAPI standard defines each event as a statement, where the required attributes are *actor*, *verb* and *object*, which correspond to user ID, action and target attributes of the interaction event described in [58]. Some of the optional attributes of the xAPI statement are *result*, *context* and *timestamp*. The structural interoperability with the xAPI standard is ensured by using the JavaScript Object Notation (JSON) format, while semantic interoperability is achieved by applying semantic rules determined by vocabularies defined for individual ar-

eas of application. The xAPI standard does not place any restrictions on the vocabulary that can be used in statements, i.e., it is possible to specify the verbs and objects to define domain-specific vocabulary [58]. It is also possible to create your own xAPI profile which contains extensions to the standard that meet new or unique needs, e.g., a vocabulary for tracking interactions in e-books or videos [58]. For example, there is a defined xAPI profile for serious games (<https://profiles.adlnet.gov/profile/0dc3dbf4-7ec9-42a2-bb3b-b9487e1b5769>, (accessed on 15 December 2024)). This makes it possible to define verbs and objects specific to serious games based on handheld AR in the context of accessibility as well.

3.1.4. Accessibility

Although previously mentioned features can affect accessibility (or reveal accessibility-related issues), in this section, we describe features and interaction parameters that can be additionally tracked. They are mostly related to the accessibility options that the user can apply or change. Changing a certain option in the accessibility menu can indicate that the user wants to improve accessibility, i.e., their user experience. Also, the context in which the change appeared needs to be recorded (e.g., the AR task or environmental context) as well as the final option with which the user ended the game.

According to research conducted regarding the needs of people with different disabilities in the context of web accessibility and mobile accessibility [64,65], the accessibility options menu should contain options to change text size, change font type, choose between different combinations of text-background colour contrast, underline and highlight links, select grayscale images, turn off light/night view, and return to initial settings. Since handheld AR has some common components as mobile applications, e.g., parts of the user interface, it is useful to define interaction parameters related to accessibility options that can be selected from the accessibility menu and to track their changes. In addition to the usual accessibility options for mobile applications, it is desirable to define and track AR-specific accessibility options, e.g., a freeze option. Other accessibility features found in AR-based solutions for people with disabilities include switching on sound support, additional visual support, wider spacing between texts, a high-contrast option for displayed 3D objects, voice control, larger manipulation buttons, sliders instead of buttons, and others [22]. Examples of accessibility options whose change in AR-based games is useful to track are shown in Table 3. They are divided into two groups of related accessibility features (UI-related and AR-specific).

Table 3. Examples of accessibility options grouped in groups of related accessibility features.

Accessibility Feature Group	Accessibility Options
User interface	Font size
	Font type
	Contrast combination
	Sound support
	Button size
AR-specific	Freeze
	High contrast
	Highlighting the edges of the object
	Buttons/sliders

3.1.5. Context

Contextual data can greatly influence the experience and content of AR due to the possibility of personalization to each user according to their location, interests and needs, especially today when devices (smartphones, tablets) are equipped with sensors that enable motion tracking and depth perception for a more comprehensive and interactive

AR experience [66]. Considering that some contextual data can change in real-time, there is a need for dynamic adaptation of interaction and content. For example, contextual data related to the circumstances of the environment in which the user uses the application (e.g., amount of light, atmospheric conditions and noise) are changed in real-time [47]. The lighting conditions affect the ability to recognize suitable AR surroundings, especially for surfaces with reflection, and to correctly display virtual objects in the real environment (non-recognition, occlusion, superimposition), especially in marker-based AR [67]. By measuring the intensity of light, it is possible to influence the colour adjustment of virtual objects, which can be useful not only for improving accessibility but also for improving the user experience of the game in general. In addition to adapting AR content to the user based on lighting conditions, it is possible to track the user's relative distance from the target (or point of interest) and adjust the size of the displayed virtual content, as well as measure ambient noise and adjust the sound level to the platform and the user's hearing abilities [44]. Besides environmental contextual data, the type of handheld device used by the user can also impact accessibility. For example, smartphones are generally smaller than tablet devices. Based on that, the user can hold and operate the device differently, e.g., with one or both hands. Also, smartphones have a smaller screen than tablet devices, which can affect the amount and type of content displayed on the screen or the layout of the buttons on the device screen.

Table 4 shows examples of contextual parameters related to the environment and device when using AR-based solutions on handheld devices. By tracking such contextual data, we can define ways of adapting the interaction and content of AR-based serious games in the context of accessibility.

Table 4. Examples of contextual parameters related to the environment and device.

Context	Examples of Parameters
Environment	Ambient intensity Ambient noise Target distance
Device	Device type Screen size

3.2. Data Model

After understanding different features that can impact the accessibility of AR-based serious games for handheld devices, it is possible to specify the user interaction tracking model in more detail. Important steps in specifying the mentioned model are the following:

1. defining the key points of interaction in AR-based serious games with triggers that initiate events, i.e., user interaction tracking. For example, *Launching a serious game* is a key interaction point in which an event is triggered when the user launches a serious game on the device;
2. specifying different interaction parameters for tracking events in the context of accessibility. These parameters can provide insight into the learning outcome and potential accessibility issues associated with using the game. Examples of parameters are *duration* (time spent in a certain game level) and *used help* (number of help requests);
3. defining the data structure of previously specified events by mapping of interaction event attributes and previously defined interaction parameters to xAPI statement attributes. Attributes are mapped as described in Section 3.1.3. while tracking of interaction parameters specified in the previous step is enabled by mapping them to *result* and *context* xAPI attributes.

Details related to each step important for specifying the model for tracking user interaction data in AR-based serious games are available in Appendix A.

The data model for tracking user interactions in AR-based serious games for mobile devices specified as previously described, relying on the xAPI standard, can be integrated into existing GLA systems. Figure 2 represents the specified data model showing how previously defined interaction parameters from the conceptual model based on features related to the user, serious game, augmented reality, context and accessibility are mapped to attribute properties in xAPI statements.

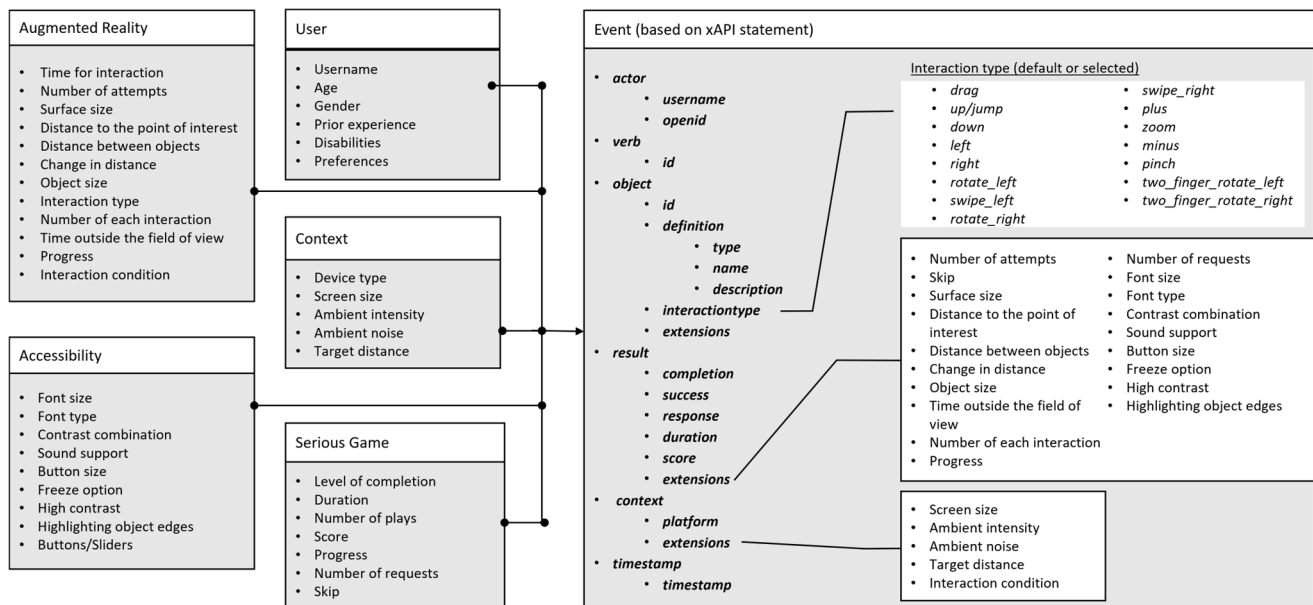


Figure 2. A data model for storing interaction parameters in an event based on the xAPI standard.

4. A Framework for Improving Accessibility of Serious Games in Handheld AR

This section describes the proposed framework for tracking user interaction data in AR-based serious games in the context of accessibility. The framework is based on the methodology for the automated collection, storage and analysis of user interaction data we will first describe. Afterwards, we explain how the framework is used in the experiment and present the results focused on improving accessibility.

4.1. Methodology for Automated Collection, Storage and Analysis of User Interactions

The goal of the proposed methodology for the automated collection, storage and analysis of interaction parameters is to enable the adaptation of user interfaces (content appearance) and interaction mechanisms in AR-based serious games for handheld devices to individual users' needs. A user interaction tracking mechanism based on the previously specified data model for tracking user interactions in the context of accessibility is implemented within this methodology. The methodology consists of four main activities as follows: (1) Understanding user needs, (2) Selection of technology and implementation of user interaction tracking, (3) Analysis of user interaction data in the context of accessibility, and (4) Implementation of adaptation of serious games. Figure 3 shows a graphic representation of the methodology in the form of a series of activities. Each of these activities is further described in more detail.

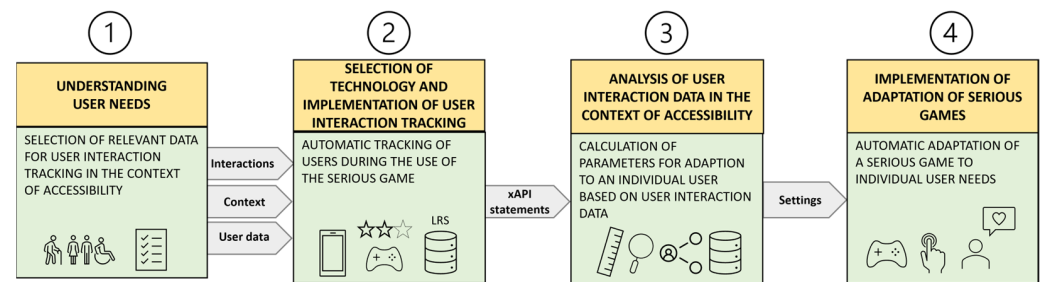


Figure 3. Graphic representation of the methodology for the automated collection, storage and analysis of user interactions in the form of a sequence of activities 1–4.

4.1.1. Understanding User Needs

The first activity of the methodology relates to understanding the accessibility issues that users of serious games based on AR for handheld devices may face, i.e., how to enable all users, regardless of their abilities or age, to use the above solutions without barriers. To understand the needs of different users, it is necessary to review the literature on studies evaluating AR-based solutions developed for people with disabilities. It is important to review the literature concerning existing guidelines or best practices for the accessibility of AR solutions, but also to examine mainstream AR applications to identify the most common tasks and interactions in AR in different application areas, including serious games in education. Also, including people with different disabilities in the process would impact a better understanding of their needs. This activity aims to identify and select relevant data (features) to be tracked while users play an AR-based serious game. This data will serve as input that will later enable the identification of potential accessibility issues or general usability problems with the serious game. When identifying the relevant data, one should have in mind the objective of tracking that data in the context of accessibility. Section 3 explains the features that need to be considered when defining the user interaction tracking model related to the accessibility of handheld AR-based serious games.

4.1.2. Selection of Technology and Implementation of User Interaction Tracking

After selecting the data to be tracked in the context of accessibility in the previous activity, it is necessary to define a mechanism for tracking user interactions. The goal of the second activity of the methodology is to enable the automatic tracking of user interactions when playing a serious game. This requires selecting the technology for implementing the software for tracking user interactions at key interaction points in a serious game and a method for distinguishing individual game sessions by users. A serious game that generates events to be tracked in the context of accessibility must be integrated with an analytics platform that supports event tracking. As the focus is on software solutions that are serious games, it is necessary to consider how different game engines can support data tracking, e.g., Unity [68] or Unreal Engine [69].

It is recommended that support for tracking interaction parameters is implemented during game development while functionality for sending events with specific interaction parameters to an analytics platform should be implemented at defined key interaction points in a serious game. The xAPI standard supports event-based tracking in serious games (as described in the previous section) whereby xAPI statements are usually stored in a Learning Record Store (LRS). This is a database with defined endpoints to which data is sent, and which stores it securely and enables further processing. Some of the well-known LRSs that support the xAPI standard are Learning Locker [70], Watershed [71], Rustici LRS [72] and Yet Analytics LRS [73]. Although the focus of LRSs is on storing and managing xAPI statements, some LRSs offer advanced analytics options. Most platforms offer basic reporting and data visualisation. If more advanced analytics options are required that

the chosen analytics platform does not offer (e.g., data modelling, predictive analytics, machine learning algorithms), other tools or platforms should be used and integrated or implemented as required.

4.1.3. Analysis of Tracked Data in the Context of Accessibility

The third activity of the methodology involves the selection of technology and implementation of software for advanced analysis of user interactions (if required) and the calculation of settings for adapting the serious game in AR to an individual user. If the selected LRS does not support advanced analysis according to the defined requirements, it is necessary to implement it on your own or to integrate an existing tool that has the necessary support. This kind of advanced analysis can help in determining parts of the game or interaction mechanisms that have the greatest impact on knowledge acquisition, accessibility and overall customer experience.

This activity aims to determine the interaction parameters (accessibility settings) that will enable the adaptation of user interfaces (content appearance) and interaction mechanisms in AR-based serious games to better suit the needs of each user in the context of accessibility. The software responsible for calculating the settings should be able to retrieve the relevant user interaction data (collected in the format of xAPI statements) from the LRS to be able to perform data analysis to determine interaction parameters for each user. Usually, LRSs provide the ability to access/query stored xAPI statements. For example, Learning Locker allows access to xAPI statements or even custom filtration of statements via defined application programming interfaces (APIs) [70]. In general, an API enables communication between different systems through the exchange of predefined messages in a format such as JSON or XML.

The implementation of software for analysing the user interaction data depends on the programming language selected for writing scripts/applications for analysing data, storing results and communicating with other applications (e.g., Python, R, Java, JavaScript, etc.). Analytical libraries or tools corresponding to the selected programming language can be used to process and analyse data from LRS. The analysis results, i.e., the settings for an individual user, must be then saved in the selected database.

4.1.4. Implementation of the Adaptation of Serious Games

The last activity of the methodology is related to the adaptation of AR-based serious games to the needs of users in the context of accessibility based on the calculated settings from the previous step. To make this possible, the calculated settings for an individual user must be made accessible to other applications in a suitable form, i.e., to the same or different AR-based serious game. Therefore, the access points, i.e., APIs, must first be implemented in the existing server part of the system to enable access to the database and retrieval of the calculated settings. Subsequently, functionality for retrieving individual user settings (API) and a mechanism for adapting selected settings in an AR-based serious game must be implemented using the selected technology. A serious game must contain a user login/registration component so that the settings can be adapted for the individual user. Adaptation of interaction mechanisms and content appearance in a serious game includes retrieving settings for an individual user by sending a request to a previously defined and implemented access point on the server side and applying defined interaction parameters from the settings in gameplay. Which settings will be applied depends on the interaction mechanisms and accessibility options related to content appearance or user interface available in certain serious games. For example, if a setting refers to the type of font, the value of that interaction parameter must be applied to all text objects that appear in a serious game.

To verify the validity of the proposed methodology, and thus the proposed model for tracking user interaction on which the methodology is based, we developed a technology platform that enables the automated collection, storage and analysis of user interaction data in the context of accessibility to adapt interaction mechanisms and content appearance in AR-based serious games for mobile devices to an individual user.

4.2. Implementation of Technology Platform

The practical application of the technology platform includes two prototypes of serious games based on AR for mobile devices. One serves as a source of user interaction data, while the other serves as a prototype of a serious game that adapts the settings of a game to the user based on the analysis of data collected. The main components of the technological platform are:

- a serious game with functionalities for collecting and sending data on user interactions and user preferences (SG1);
- an analytics platform that stores data sent by a serious game and enables basic visualisation and analysis of the results;
- software for advanced analysis of selected user interaction data and calculation of settings for an individual user stored in the database;
- a serious game with an implemented mechanism for adapting selected settings (UI parameters and interaction mechanisms) for an individual user based on retrieved settings data from the database (SG2).

The components are shown in Figure 4 in the order from left to right in which they are described. The implementation of each component corresponds to the activities of the proposed methodology and the technology used for each component of the technology platform is described below.

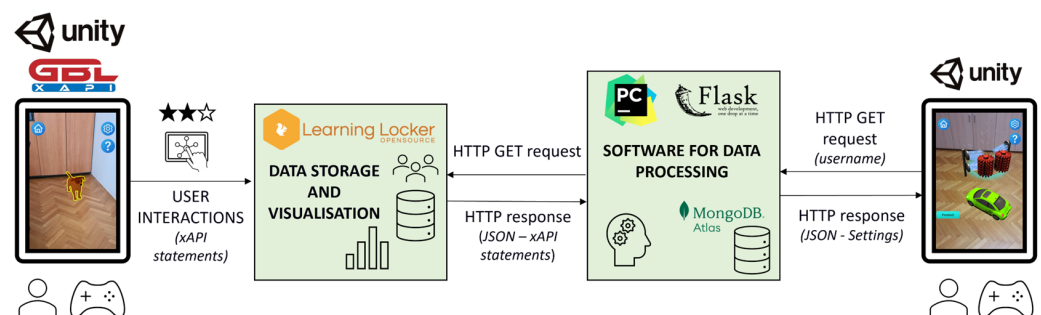


Figure 4. Graphical representation of the components of the developed technology platform.

4.2.1. Serious Game SG1

A serious game SG1 was developed using the Unity game engine [68] which has support for AR development. In particular, the AR Foundation package (<https://docs.unity3d.com/Packages/com.unity.xr.arfoundation@3.1/manual/> (accessed on 15 December 2024)) was used to develop the AR features. This package enables the development of multi-platform AR applications with Unity. For the development of this serious game, the ARCore XR Plugin (<https://docs.unity3d.com/Packages/com.unity.xr.arcore@4.1/manual/index.html> (accessed on 15 December 2024)) was used to enable the development of AR for Android devices.

The SG1 contains nine tasks in which the user has the choice between different interaction mechanisms or has to complete the task with a specific interaction mechanism. To start the game, it is necessary to first scan the surface on which virtual objects are then created. The main character of the serious game is a dog with whom the user must perform different interactions depending on the task. If a user does not know the task or cannot solve it, it is

possible to skip the task, with the *Skip* button appearing 10 seconds after the task starts, or the *Help* button starts blinking to give the user hints on how to perform the task. During the game, it is also possible to change the accessibility settings related to font type and text size in the game. Table 5 brings the description of nine tasks with possible interaction mechanisms in it. As one can notice, tasks related to the virtual object's transformation (translation—tasks 2 and 3, scaling—tasks 4 and 5 and rotation—tasks 7 and 8) offer only one type of interaction mechanism (touch gesture or indirect controls, i.e., buttons), while others offer a choice (besides task 1 where an object needs to be activated with a tap gesture). After finishing the last task, the user has the option to give subjective ratings of satisfaction with individual interaction mechanisms which can be considered when calculating the settings related to interaction. Figure 5 shows screenshots from SG1 where tasks with different types of interaction mechanisms are shown.

Table 5. SG1 tasks and interaction mechanisms.

#	Task	Interaction Mechanism
1	Select the virtual object (dog)	Tap gesture
2	Translate the virtual object (dog)	Drag and drop gesture
3	Translate the virtual object (dog)	Arrow buttons (up, down, left, right)
4	Scale the virtual object (biscuit)	Pinch-to-zoom gesture
5	Scale the virtual object (biscuit)	Resize buttons (plus, minus)
6	Translate the virtual object (dog)	Drag and drop gestures or arrow buttons (up, down, left, right)
7	Rotate the virtual object (dog)	Rotation gesture (swipe or two-finger-rotate)
8	Rotate the virtual object (dog)	Rotation buttons (for left and right rotation)
9	Enter the numbers from the virtual object (dog's collar) into the input field	All interaction mechanisms available (touch gestures and indirect controls)

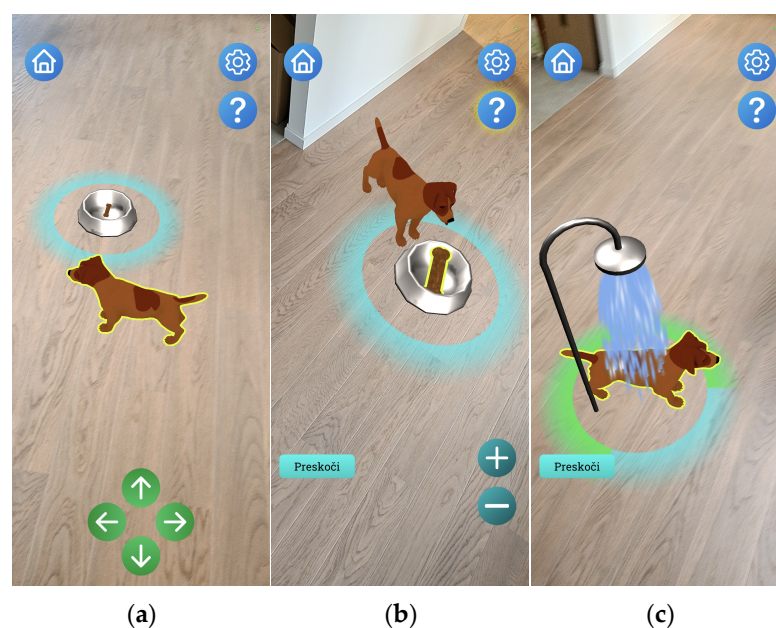


Figure 5. Screenshots from SG1: (a) translation task with buttons, (b) scaling task with buttons, (c) rotating task with gestures.

Support for tracking user interaction data at key interaction points has been implemented in SG1. The interaction parameters for SG1 are defined according to the interaction parameters for tracking events in the context of accessibility, which are described in Section 3.2. Then, the SG1 interaction parameters and their values are mapped to the xAPI attributes accordingly, while the support for sending xAPI statements from SG1 is implemented using the GBLxAPI (<https://github.com/gblxapi/UnityGBLxAPI> (accessed on 15 December 2024)) library for Unity. In addition to these interaction parameters which represent objective metrics, subjective metrics, i.e., ratings of interaction mechanisms used across tasks, are also sent as part of the xAPI statements from SG1.

4.2.2. Learning Locker LRS

The LRS integrated with the technology platform and responsible for storing the xAPI statements from the SG1 is Learning Locker [70]. It is an open-source scalable database, free to use (with limited functionalities) and it can be custom-installed on a virtual machine with certain requirements (<https://learninglocker.atlassian.net/wiki/spaces/DOCS/pages/106102785/Custom+Installation> (accessed on 15 December 2024)). Learning Locker provides many options for retrieving xAPI statements. To satisfy our needs, we used the xAPI HTTP interface (<https://learninglocker.atlassian.net/wiki/spaces/DOCS/pages/106496109/xAPI> (accessed on 15 December 2024)) for retrieving all stored xAPI statements from SG1, and the Aggregation API (<https://learninglocker.atlassian.net/wiki/spaces/DOCS/pages/106037259/Aggregation+API> (accessed on 15 December 2024)) for retrieving statements filtered by the user. Since Learning Locker does not provide the advanced analysis of user interactions in the context of accessibility defined in this paper, we implemented software support for retrieving and processing xAPI statements with user interaction data aimed at calculating the parameters of accessibility settings for an individual user.

4.2.3. Software for Data Processing

Software support for advanced analysis of selected user interaction data and calculation of settings for an individual user stored in the database consists of a web server developed in Flask, a development environment for the Python programming language, and a MongoDB database. The calculated settings for each user are stored in the database, where the user is distinguished by the username that the user enters when logging into the serious game SG1. The settings selected for the adaptation in the prototype of a serious game SG2 are described in Table 6. The table contains a description of each setting and its values. The values are grouped into default and accessible adaptation options. SG1 contains tasks with interaction mechanisms corresponding to Translation, Rotation and Scaling settings from the table. Also, in SG1, the user can change options related to font/text, which correspond to the Font Type and Text Size settings from the table. Each of the listed settings has a default and alternative option that is implemented in the game. The method used for calculating the settings for an individual user is described below.

Table 6. Selected settings to adapt the serious game to the user.

Settings Group	Setting Name	Setting Description	Adaptation Options	
			Default	Accessible
UI	Font type (<i>textFont</i>)	The font type applied to all text objects in the game	Roboto	OpenDyslexic
	Text size (<i>textScale</i>)	Text size applied to all text objects in the game	1	>1, max 1.5

Table 6. Cont.

Settings Group	Setting Name	Setting Description	Adaptation Options	
			Default	Accessible
Interaction mechanism	Selection (<i>tapStyle</i>)	Interaction type for selection (of a virtual object or an option that affects a virtual object)	Tap gesture	Select button
	Translation (<i>movingStyle</i>)	Interaction type for moving a virtual object from one place to another	Drag and drop gesture	Arrow buttons (up, down, left, right)
	Rotation (<i>rotationStyle</i>)	Interaction type for rotating the virtual object in two directions (left/right)	Rotation gestures (swipe, two-finger-rotate)	Left/right rotation buttons
	Scaling (<i>scaleStyle</i>)	Interaction type for changing the size of the virtual object (reduce/increase)	Pinch-to-zoom gesture	Resize buttons (plus, minus)

4.2.4. Calculation of Settings for an Individual User

By tracking user interactions according to specified interaction parameters, it is possible to extract metrics from a set of xAPI statements that can help determine which settings better suit an individual user in the context of accessibility. A selected set of metrics is taken into account to calculate the interaction mechanism option in each of the tasks for transforming the virtual object. The metrics considered are related to:

- skipping the task—parameter *skipped* in the *extensions* property of the *result* attribute for the verb *completed* and the object type *level* (task);
- rating of the interaction mechanism—parameter *score* of the *result* attribute for the verb *assessed* and the object type *interaction*;
- task completion time—parameter *duration* of the *result* attribute for the verb *completed* and the object type *level* (task).

Since the usual interaction mechanisms in AR for mobile devices are touch gestures, the user is also offered a touch gesture as a default interaction option here. If the default interaction appears inappropriate for the user, an alternative option is offered (indirect control). The method for determining the interaction mechanism for a particular task is illustrated in Figure 6 using a UML activity diagram. If the user skips one of the related tasks (related tasks are tasks that have the same object transformation goal: translation, rotation or scaling), then the interaction mechanism of the related task that is not skipped is assigned to the user. For example, if a task with touch gestures is skipped, the setting for this task will be indirect controls. If the user skips both related tasks, the interaction mechanism in the settings will be the default one. If the user does not skip any related tasks, the user's subjective rating for the interaction mechanisms will be considered. If there is an interaction mechanism that is rated higher, the interaction mechanism from that task will be assigned to the user. If the interactions of related tasks are rated equally or at least one interaction is not rated, the completion time of the task is considered, with the task that is completed faster taking priority. If related tasks have the same completion time, the setting for this task will be the default interaction mechanism.

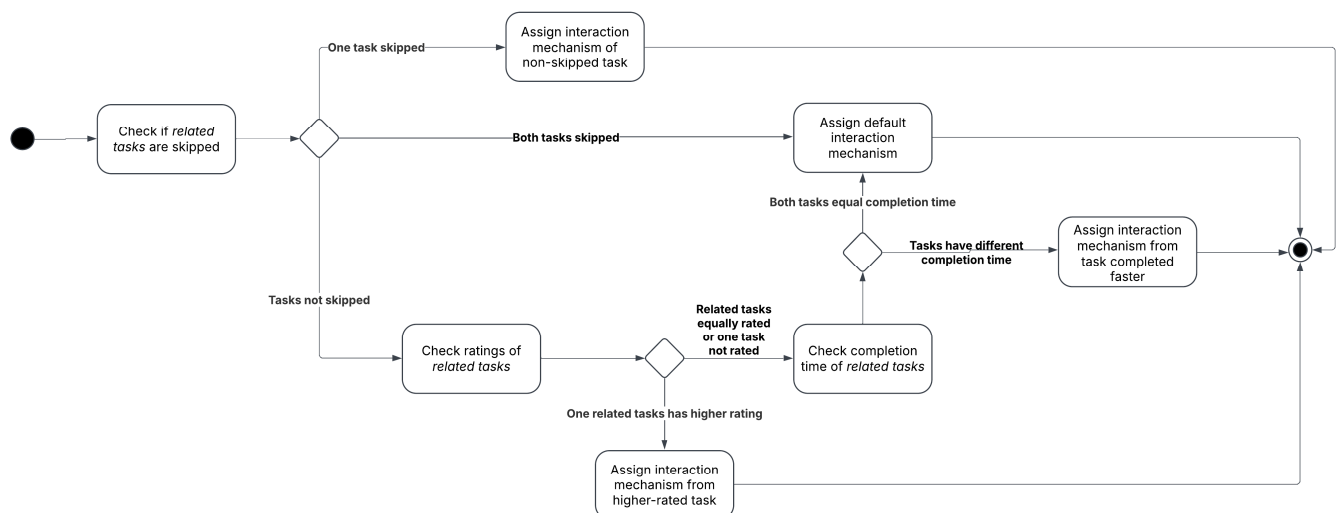


Figure 6. UML activity diagram illustrating a method for determining the interaction mechanism in related tasks.

The Selection task does not have its pair in the form of related tasks. However, it is assumed that the interaction mechanism calculated for the Translation task is also suitable for the Selection task. Thus, if the user's setting for the Translation task is the touch gesture, then for the Selection task, the setting is the tap gesture (because the drag and drop gesture contains the elements of object selection). Otherwise, an indirect control (a Select button) is used as the setting. As for the adaptation options for the UI settings, the options with which the user has finished the game are considered UI settings. It is assumed that the user has finished the game with the most appropriate settings, as there is the possibility to change them in the game.

4.2.5. Serious Game SG2

SG2 is a prototype game with an implemented mechanism for adapting selected settings (UI parameters and interaction mechanisms) for an individual user based on retrieved settings data from the database. It is developed in Unity and its functionalities are similar to SG1. It comprises six tasks in which the user is given the interaction mechanism according to the calculated settings. The settings for an individual user are accessed by sending a request to a defined access point on the server part of the system. The response to the request contains the settings values in JSON format. The main character of SG2 is a car with which the user needs to perform different actions depending on the task. As in SG1, skipping the task and changing the UI settings is possible. Before entering the game, the user logs in with a username, after which the settings for that user are retrieved. If a user did not previously play a full session of SG1, they will play SG2 with the default settings. The user first starts a game by scanning the surface on which the virtual objects are then created. Afterwards, the user performs the tasks with assigned interaction mechanisms (touch gesture or indirect control). The six tasks of SG2 are related to changing the car's colour, translating, scaling and rotating the car, and entering the number from the car registration in the input field. Figure 7 shows screenshots from SG2 where tasks with interaction mechanisms assigned for a test user are shown.

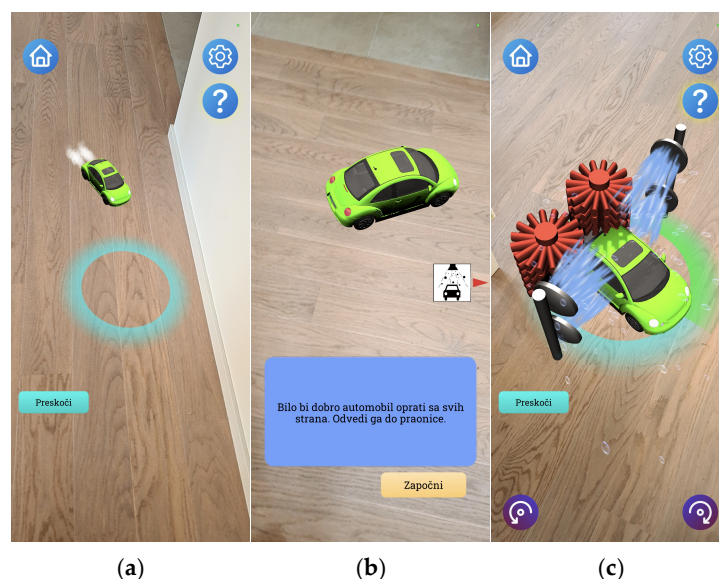


Figure 7. Screenshots from SG2: (a) translation task with gestures, (b) instruction for translation after scaling the car, (c) rotating task with buttons.

By having implementation of the next three modules: module to support automated tracking of user interaction data, module to analyse user interaction data and calculate adaptation settings for an individual user, and module to support adaptation of interaction mechanisms in selected tasks and UI parameters in serious games in AR, key parts of the framework that enable the goal of improving accessibility of AR-based serious games are developed. The framework in its entirety is shown in Figure 8. To determine the theoretical and practical validity of the framework proposed in this paper, an experiment with a developed technology platform was conducted. The experiment included user testing with people with disabilities and people of different ages.

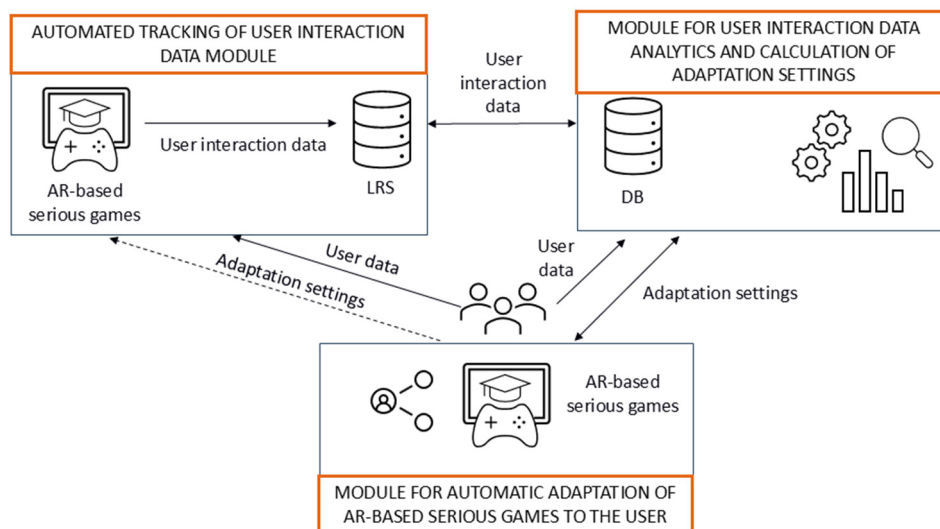


Figure 8. A framework for tracking user interactions for improving accessibility of serious games in handheld AR.

5. Experiment Results and Discussion

The technology platform based on the proposed methodology and model was evaluated by conducting user testing with people with disabilities and people of different ages. This section describes the user testing and analysis of the experiment's results.

5.1. Experiment Description

The user testing was conducted with a total of 108 participants, and according to the universal design, the group of participants included people with disabilities and people of different ages. Specifically, 29 people with different disabilities (visual, auditory, physical, cognitive/neurological, specific learning difficulties, multiple disabilities) and 79 people of different ages participated in the user testing. Most of the users with disabilities were recruited in agreement with associations of people with various types of disabilities from Zagreb (Croatia), while to recruit other participants, the study was disseminated among UNIZG FER staff and their contacts.

According to research from [74], five participants are sufficient for testing the usability of a solution, as you can find as many usability problems with five users as with many more users, which is not always cost-effective. However, the formula from [74] is valid when it comes to users who use the product in a similar way. When it comes to different groups of users, such as people with different types of disabilities or people with different ages, then it is necessary to test the product with users from each group. Although the user groups are different, there will be some overlap in the findings of errors, so it is recommended that at least three users from each user group be involved if there are three or more user groups [75]. Therefore, in recruiting participants for user testing in this research, care was taken to collect at least the minimum number of participants of people with different types of disabilities and people of different age groups necessary to find relevant usability and/or accessibility problems for each group.

The experiment occurred in the spring of 2023 (from April 20 to May 9) in different environments and cities in Croatia, where participants completed the tasks in a real-world setting. The largest proportion of participants tested in the Laboratory for Assistive Technologies and Alternative and Augmentative Communication (ICT-AAC) (<http://lab.ict-aac.hr/> (accessed on 15 December 2024)) at the University of Zagreb Faculty of Electrical Engineering and Computing (UNIZG FER), while most of the remaining testing locations included the premises of associations for people with different types of disabilities. Two models of Samsung tablet devices (Galaxy Tab S6 Lite, 4/64 GB and Galaxy Tab A8, 3/32 GB) were used for testing the games and filling out the questionnaire.

Before participating in the user testing, each participant filled out and signed an informed consent form, which described the scope of the research in which the user testing was conducted, what data was collected, the testing procedure, and how the privacy of each user was ensured. For participants who could not sign the form by themselves, parents or caregivers did that for them. User testing was performed by the main researchers (authors in the paper) with the help of two UNIZG FER students. In the associations with people with disabilities, the participants first listened to a short, 10-min lecture on AR with a demonstration of various applications on tablet devices. The terminology used was adapted so that technological terms are brought closer to people with certain disabilities.

The main activities of the user testing included playing the first AR-based serious game (SG1) on a tablet device, then playing the second AR-based serious game (SG2) on a tablet device and filling out the questionnaire created in Microsoft Forms. The total duration of user testing was estimated at 10–15 min per subject, except for testing with participants with certain types of disabilities, which lasted from 20 to 30 min. Additionally, after filling out the questionnaire, a discussion with individual users with disabilities and/or their parents/assistants took place regarding certain accessibility problems and in that case, the entire user testing took longer (up to 45 min). Each participant was assigned a unique username, which relates to the user interaction data collected during the play of SG1. By this username, the settings for each participant for SG2 were then retrieved. Participants also used this username when filling out the questionnaire.

The purpose of the first user testing activity is to collect data on user interactions and ratings for the interaction mechanisms. SG1 tracks user interaction data and sends them to the database (LRS) in real-time (as described in Section 4.2.1). The selected set of interaction data is then used to calculate the accessibility settings in the data processing part of the system (as described in Section 4.2.4). After that, SG2 retrieves the settings and adapts the game accordingly (as described in Section 4.2.5). During user testing, devices were connected to the Internet to store and retrieve the data. The questionnaire aims to collect qualitative data related to SG2 to evaluate its usability/accessibility, satisfaction with interactions and general impression of the game. Additionally, tasks with assigned interaction mechanisms are also assessed to compare the ease of use to the first game where users experienced both interaction mechanisms. When it was easier for the participant, the questionnaire was accessed on paper. In addition to questions about the usability, accessibility and user experience of SG2, the questionnaire contained questions to collect basic demographic data about the user. A complete questionnaire with questions and possible options for answering is available in Appendix B.

Table 7 shows the distribution of participants by age group and disability type, i.e., the absolute and relative frequency of occurrence of a particular age group and disability. Among participants with disabilities, the most represented is the category of participants with multiple disabilities in which all participants have a physical disability in combination with other types of disabilities, i.e., cognitive and neurological disorders/specific learning disabilities, visual impairments and auditory impairments. Although 19 participants answered that they had some type of visual impairment in the questionnaire, we did not consider those participants whose answers did not fall under the definition of visual impairment as defined by the World Health Organization (WHO) [76]. For this reason, the recommended number of participants in usability testing according to Nielsen [75] was not met for this category, which represents one of the limitations of this study.

Table 7. Distribution of participants by age and disability type.

Age Group	f (N _a = 79)	%	Disability Group	f (N _d = 29)	%
Under 18	9	11.39	Visual	1	3.45
18–24	37	46.84	Auditory	3	10.34
25–34	12	15.19	Physical	5	17.24
35–44	6	7.59	Cognitive, learning, neurological	8	27.59
45–54	7	8.86	Multiple	12	41.38
55–64	5	6.33			
65 and more	3	3.80			

5.2. Quantitative Data Analysis

Based on the interaction parameters defined for tracking events in the context of accessibility in serious games, quantitative data were collected and stored in LRS. We exported data from Learning Locker and analysed interaction parameters important for determining the interaction mechanism settings for individual use, such as task completion time, the option to skip tasks, and the subjective assessment of interactions. The focus of results analysis is on overall descriptive trends for these parameters rather than on comparing specific participant categories or establishing statistical relationships. Therefore, inferential analyses were not necessary. Instead, we employed descriptive statistics to summarize key findings across all participants.

5.2.1. Completion Time Task

Descriptive statistics were calculated for the time required to complete each transformation task (Translation, Scaling and Rotation) with two interaction mechanisms (Touch Gestures and Indirect Controls). Calculated measures, i.e., the mean (M), median (C), standard deviation (Sd), minimum (min) and maximum (max) value, first and third quartiles (Q1 and Q3) and 95% confidence interval (CI) for each transformation task and interaction mechanism are shown in Table 8. The time per task for all participants is also shown graphically using a box and whisker plot in Figure 9 with the following values marked for each task: the maximum, minimum, interquartile range (IQR), mean (M), median (C), and outliers for each task. The data represent the time required by all participants to complete the tasks with the respective interaction mechanisms, as summarized in Table 8.

Table 8. Descriptive statistics for the time required to complete each transformation task (Translation, Scaling, and Rotation) with two interaction mechanisms (Touch Gestures and Indirect Controls). Time is reported in seconds. TG = Touch Gestures, IC = Indirect Controls.

	Translation (TG)	Translation (IC)	Scaling (TG)	Scaling (IC)	Rotation (TG)	Rotation (IC)
M	9.78	10.86	14.04	7.69	16.20	5.94
C	4.83	5.85	11.26	5.50	10.82	4.10
Sd	11.96	14.13	13.41	9.17	15.12	8.11
min	0.73	1.30	2.56	1.07	0.73	1.60
max	68.07	82.4	87.83	61.44	86.13	80.53
Q1	2.77	3.28	5.72	3.16	6.15	2.7
Q3	11.55	12.4	16.77	8.01	23.11	6.53
CI = 95%	7.52; 12.03	8.19; 13.52	11.51; 16.57	5.96; 9.42	13.35; 19.05	4.41; 7.47

Note: M—mean, C—median, Sd—standard deviation, min—minimum, max—maximum, Q1—first quartile, Q3—third quartile, CI = 95%—95% confidence interval

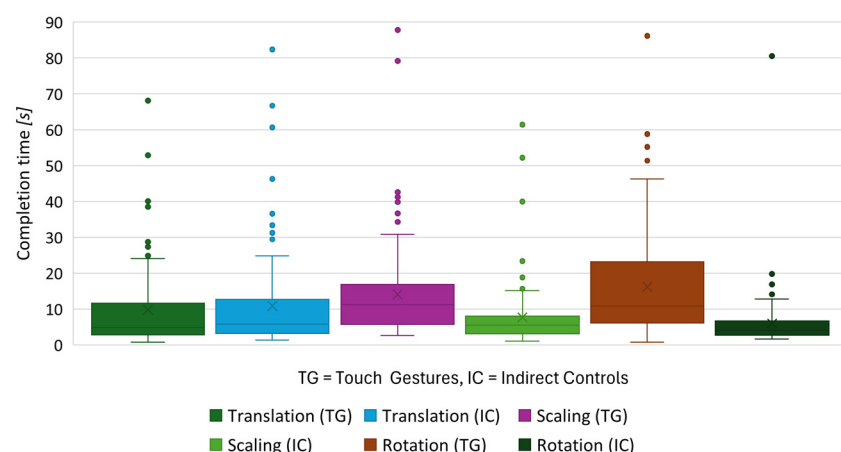


Figure 9. Box and whisker plot illustrating the distribution of completion times (in seconds) for each transformation task (Translation, Scaling, and Rotation) with two interaction mechanisms (Touch Gestures and Indirect Controls).

The Scaling (TG) and Rotation (TG) tasks stand out, as they have average and median values higher than the remaining tasks. Both tasks include touch gestures. However, in scaling with touch gestures task, the user first needs to select the object before performing the required interaction with it, so it is possible that some participants did not immediately notice this. In rotation with touch gestures, two modes of interactions for rotation are possible (swipe and rotate with two fingers). In this task, the user should rotate the object

for a full circle, which is indicated by a circle that fills as the user rotates. More time to solve this task may therefore also result from the fact that the indicator prompted the user to make a swipe gesture in a circle, which is not recognized as the correct interaction mechanism. The Rotation (IC) and Scaling (IC) tasks have the lowest average and median values indicating that participants completed these tasks faster than the remaining tasks. The diagrams for all tasks show a positively asymmetric distribution of data.

According to the calculated descriptive statistics and from a look at the previous graph, it can be concluded that the values of completion time for tasks with touch gestures for scaling and rotating the object are higher than their related task, i.e., scaling and rotating the object with indirect controls. This result shows that the choice of interaction mechanism for scaling and rotation tasks impacts task performance time. Therefore, scaling and rotation tasks in AR can be considered easier for most of the participants when they are performed with indirect controls than with touch gestures.

5.2.2. Skipped Tasks

Figure 10 shows a pie chart presenting the count and percentage of participants who skipped each transformation task. The task that was skipped the most was the Rotation (TG) task (26 times, i.e., 24.07% of participants skipped it). The next task is the Scaling (TG) task, which was skipped 22 times (20.37% of participants skipped it). The Translation (IC) task was skipped 8 times (7.41% of participants skipped it), the Translation (TG) task was skipped 7 times (6.48% of participants skipped it), and the Rotation (IC) task was skipped 4 times (3.7% of participants skipped it). The Scaling (IC) task was skipped only twice (2.90% of participants skipped it).

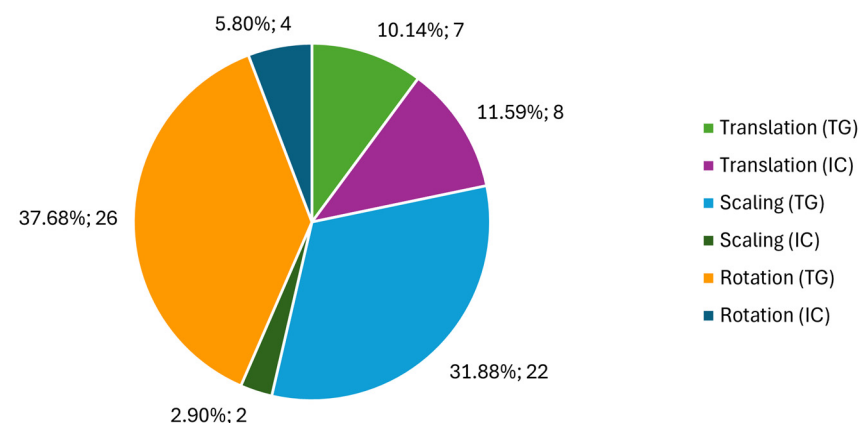


Figure 10. Pie chart showing the distribution of skipped transformation tasks (Translation, Scaling, and Rotation) across two interaction mechanisms (Touch Gestures and Indirect Controls). TG = Touch Gestures, IC = Indirect Controls.

From the results, it can be concluded that there is a higher proportion of skipped tasks in pairs of related tasks Scaling TG-IC and Rotation TG-IC in those tasks that implement touch gestures compared to those that implement indirect controls, while for the related tasks Translation TG-IC, there is an equal number of skipped tasks regardless of the interaction mechanism.

5.2.3. Subjective Evaluation of Interaction Mechanisms in SG1

Subjective evaluation of interaction satisfaction is integrated into SG1. After bringing the game to an end, interaction mechanisms are displayed on the screen, for which it is possible to give a rating by choosing the number of stars from 1 to 5, where 1 = Very dissatisfied, 2 = Dissatisfied, 3 = Neither dissatisfied nor satisfied, 4 = Satisfied, 5 = Very satisfied. User satisfaction with interactions in this context is defined as the degree to which the user's needs, expectations and preferences are satisfied by the interaction, with an

emphasis on ease of use as a factor that affects interaction satisfaction. Each interaction is supported by a visual representation to make it easier for the participants to recognize which interaction was in which task. In addition, for some participants with disabilities, the evaluation of interactions was explained orally so that their ratings would more reliably reflect the ease of use of a particular interaction.

For the analysis of the subjective evaluation of interaction mechanisms in transformation tasks in SG1, descriptive statistics were calculated (N = 107). A participant who did not rate at least one of the interaction mechanisms was excluded from the analysis. The calculated measures (average value M, median C, standard deviation Sd, minimum and maximum value, and confidence interval CI = 95%) for participants' ratings of interaction mechanisms in each transformation task are shown in Table 9. Figure 11 shows a graph with mean values of ratings for each transformation task (Translation, Scaling, and Rotation), along with error bars indicating the 95% confidence intervals. These results provide insight into participants' subjective evaluation of the interaction mechanisms, complementing the descriptive statistics presented in Table 9.

Table 9. Descriptive statistics for the subjective evaluation of interaction mechanisms in transformation tasks in SG1 (N = 107). TG = Touch Gestures, IC = Indirect Controls.

	Translation (TG)	Translation (IC)	Scaling (TG)	Scaling (IC)	Rotation (TG)	Rotation (IC)
M	4.5	4.48	4.00	4.79	3.83	4.72
C	5	5	4	5	4	5
Sd	0.9	0.95	1.2	0.69	1.31	0.74
min	1	1	1	1	1	2
max	5	5	5	5	5	5
CI = 95%	4.33; 4.68	4.3; 4.66	3.77; 4.23	4.65; 4.92	3.58; 4.08	4.58; 4.86

Note: M—mean, C—median, Sd—standard deviation, min—minimum, max—maximum, CI = 95%—95% confidence interval

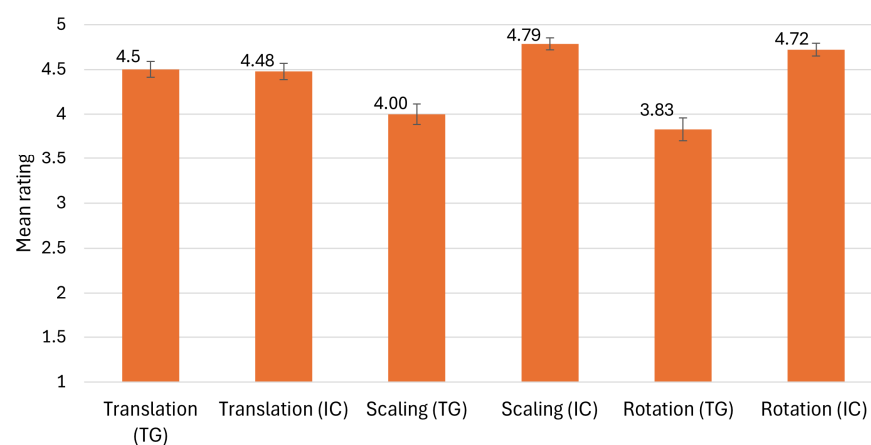


Figure 11. Mean ratings of interaction mechanisms (TG = Touch Gestures, IC = Indirect Controls) for each transformation task (Translation, Scaling, Rotation) in SG1 with 95% confidence intervals.

Results indicate that ratings for most of the tasks were distributed around rating 5, i.e., the participants were most often very satisfied with the interaction in the task. The exceptions are Scaling (TG) and Rotation (TG) tasks where the most common rating was 4, meaning that they were less satisfied with their interaction mechanisms than in other tasks. In these tasks, there is also greater variability in ratings compared to other tasks. The results correspond with the results of completion time and skipped task option, where

Scaling (TG) and Rotation (TG) tasks also stand out, so we can conclude that scaling and rotation with touch gestures are more critical tasks in SG1 than others.

5.3. Analysis of the Calculated Settings for Individual Users

In this paper, special emphasis is placed on the analysis of the calculated settings for each user, that is, the interaction mechanisms adapted to them in SG2 based on objective and subjective measures collected by tracking user interactions during the play of SG1. Since users rated their satisfaction with interaction mechanisms in both serious games, SG1 and SG2, it is possible to compare the results and user satisfaction with them. The aim of the result analysis performed as a part of this paper is to show that tracking user behaviour in terms of their performance and interaction during gameplay can be useful in revealing potential accessibility issues as well as have a positive impact on the user experience by adapting critical elements of the game. Critical elements in this case are types of interaction mechanisms used for tasks in an AR-based serious game related to the virtual object transformation.

In this section, the distribution of interaction mechanisms assigned to participants is analysed to explore how interaction mechanisms are assigned based on collected metrics rather than formally comparing groups or making predictive inferences. Therefore, descriptive statistics were used to present distributions and trends without applying inferential tests. In addition, while we present the distribution of interaction mechanisms per participant group, some subgroups have relatively small sample sizes, which may limit the reliability of inferential statistical tests.

The method for determining the interaction mechanism for a specific task of transforming a virtual object is described in Section 4.2.4. Figure 12 shows the distribution of interaction mechanisms assigned to the participants ($N = 108$). For the *Translation* task, both interaction mechanisms (touch gestures and indirect control) are equally represented (indirect control is assigned to 51.85% (56) of the participants, while touch gestures are assigned to 48.15% (52) of the participants). For the *Rotation* and *Scaling* tasks, however, indirect control is assigned to a significantly higher proportion of participants than touch gestures. More specifically, 85.19% (92) of the participants are assigned indirect control for the *Rotation* task, while the rest of the participants, i.e., 14.81% (16), are assigned touch gestures. Similarly, for the *Scaling* task, 83.33% (90) of the participants are assigned indirect control, and 16.67% (18) of the participants are assigned touch gestures.

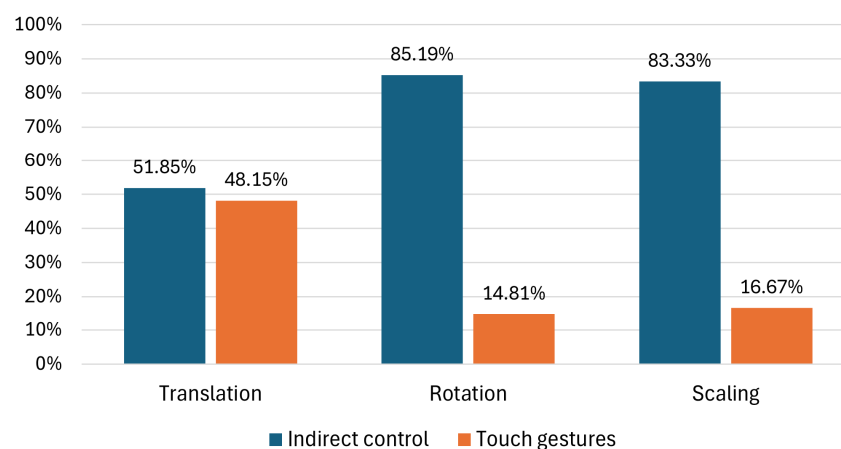


Figure 12. Distribution of interaction mechanisms assigned to participants ($N = 108$) for transformation tasks (Translation, Scaling, and Rotation). The figure shows the proportion of participants assigned to each interaction mechanism: Touch Gestures and Indirect Control.

Results of applied interaction mechanisms in SG2 for tasks of translation, rotation and scaling the virtual object for participants by disability and age groups are summarized in Table 10. More elaboration on the results for each task follows below.

Table 10. Summary of applied interaction mechanisms in SG2 for virtual object transformation tasks (Translation, Rotation, and Scaling) across different disability and age groups. The table presents the relative frequency (%) of participants assigned to each interaction mechanism (TG = Touch Gestures, IC = Indirect Controls) for each task.

	Participants Group	Translation		Rotation		Scaling	
		TG	IC	TG	IC	TG	IC
Disability group ($N_d = 29$)	Auditory ($N_a = 3$)	100	0	33.3	66.7	0	100
	Physical ($N_p = 5$)	40	60	0	100	20	80
	Cognitive, learning, neurological ($N_{cln} = 8$)	37.5	62.5	0	100	25	75
	Multiple ($N_m = 12$)	41.7	58.3	0	100	8.3	91.7
Age group ($N_{age} = 79$)	Under 18 ($N_{<18} = 9$)	33.3	66.7	0	100	0	100
	18–24 ($N_{18-24} = 37$)	44.7	55.3	18.4	81.6	18.4	81.6
	25–34 ($N_{25-34} = 12$)	58.3	41.7	33.3	66.7	8.3	91.7
	35–44 ($N_{35-44} = 6$)	83.3	16.7	16.7	83.3	50	50
	45–54 ($N_{45-54} = 7$)	28.6	71.4	0	100	0	100
	55–64 ($N_{55-64} = 5$)	40	60	40	60	40	60
	65 and more ($N_{65+} = 3$)	66.7	33.3	0	100	0	100

5.3.1. Translation Interaction Mechanisms

One visually impaired participant was assigned indirect controls for all virtual object transformation tasks (it is not shown in the table). All participants with auditory impairments were assigned touch gestures for the *Translation* task. Participants from the remaining disability groups have an approximately equal division for the *Translation* task, where a slightly larger share of participants was assigned indirect control (60% of the participants with physical disabilities, 62.5% with cognitive and neurological disorders and/or specific learning disabilities, and 58.3% with multiple disabilities).

Participants from most age groups were assigned the touch gesture, i.e., for the groups 25–34 years old (58.3%), 35–44 years old (83.3%) and 65 or more years old (66.7%), while the participants from the remaining groups were assigned indirect controls for *Translation* task, namely the groups Under 18 years old (66.7%), 18–24 years old (55.3%), 45–54 years old (71.4%) and 55–64 years old (60%).

5.3.2. Rotation Interaction Mechanisms

All participants from the physical, cognitive and neurological disorders and/or specific learning difficulties and multiple disabilities groups were assigned indirect controls for the *Rotation* task, while participants from the auditory impairments group were mostly assigned indirect control (66.7%).

The age groups Under 18, 45–54, and 65 or older stand out, where all participants were assigned indirect controls as an interaction mechanism for the *Rotation* task. Participants from the remaining age groups were mostly assigned indirect controls, i.e., 81.6% of participants from the 18–24 years old group, 66.7% of participants in the 25–34 years old group, 83.3% of participants in the 35–44 years old group, and 60% of participants from the 55–64 years old group.

5.3.3. Scaling Interaction Mechanisms

All participants with auditory impairments were assigned indirect controls for the *Scaling* task. Participants with physical disabilities were mostly assigned indirect controls (80%), as were participants with cognitive and neurological disorders and/or specific learning disabilities (75%) and multiple disabilities (91.7%).

Similar to results for the *Rotation* task, all participants from age groups Under 18, 45–54, and 65 or older were assigned indirect controls for the *Scaling* task. Most of the participants from age groups 25–34 years were also assigned indirect controls, i.e., 91.7% of them. Participants in the age group 18–24 were mostly assigned indirect controls (81.6% of them), as well as participants in the age group 55–64 (60% of them). This is the same proportion of assigned interaction mechanism as these groups have for the rotation task. Both interaction mechanisms were assigned to an equal share of participants in the 35–44 age group (50% touch gesture, 50% indirect controls).

In general, the results show that the *Translation* task stands out among the tasks concerning the assignment of interaction mechanisms. Looking at the results for all participants, both interaction mechanisms (touch gestures and indirect controls) were assigned in equal proportion to the *Translation* task. Observing the assigned interaction mechanisms for this task for different disability groups, most participants from each group were assigned indirect controls as an interaction mechanism (besides one participant from the visual impairment group). As far as different age groups are concerned, the 35–44 years group stands out in which most participants were assigned touch gestures. In general, for the *Rotation* and *Scaling* tasks, a significantly higher proportion of participants were assigned the interaction mechanism of indirect controls. In the context of age groups, the group 35–44 years stands out again, in which indirect controls and touch gestures were assigned to the participants in equal proportions for the *Scaling* task.

These results suggest that the tasks of *Rotation* and *Scaling* using the touch gesture interaction mechanisms were more difficult than those using indirect controls for most of the participants, while the *Translation* task had a similar proportion of participants finding each interaction mechanism easy / difficult.

5.4. Subjective Evaluation of Interaction Mechanisms in SG2

Apart from the questions related to other qualitative data related to SG2, the questionnaire contained questions whose aim was to examine the ease of use of the assigned interaction mechanisms. These questions were formed using simple vocabulary to be more comprehensible to people with intellectual disabilities or people with no previous experience with AR. In this sense, instead of choosing the degree of agreement with the statement, e.g., “I find the interaction in the virtual object translation task easy to use”, participants were asked to rate how difficult or easy it was for them to do what the goal of the specific task was, e.g., move the object to the required place. So, the participants answered the questions by choosing a rating from 1 to 5 where 1 = Very difficult, 2 = Difficult, 3 = Neither difficult nor easy, 4 = Easy, and 5 = Very easy. The analysis of subjective ratings aimed to provide a general overview of how participants perceived the ease of interaction with the assigned mechanisms rather than to compare differences between groups or draw inferential conclusions. Therefore, descriptive statistics were employed to summarize key findings.

First, the distribution of participants’ ratings for three tasks of virtual object transformation is calculated and presented in Figures 13–15. For the *Translation* task, most of the participants answered that it was very easy (54.63%) or easy (31.48%) to move the virtual object to the required location with the assigned interaction (Figure 13). While 9.26% of participants answered that it was neither difficult nor easy for them to solve the given task,

only 4.63% of the participants answered that it was very difficult or difficult to move the object. Among the participants who gave a rating of 3 or lower, the majority are people without disabilities, members of the age groups 18–34 years old and older than 55 years.

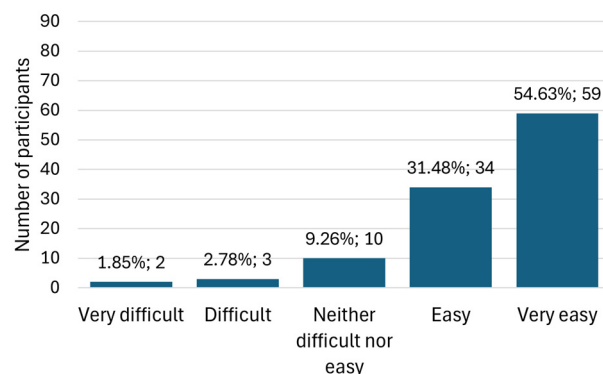


Figure 13. Distribution of participants' ratings for the ease of interaction in the *Translation* task.

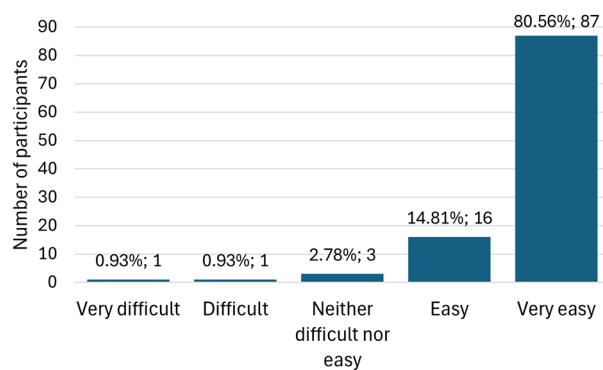


Figure 14. Distribution of participants' ratings for the ease of interaction in the *Scaling* task.

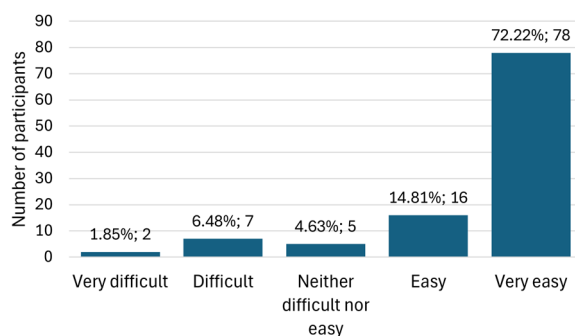


Figure 15. Distribution of participants' ratings for the ease of interaction in the *Rotation* task.

For the *Scaling* task, most participants answered that it was very easy (80.56%) or easy (14.81%), as can be seen in Figure 14. While 2.78% of participants answered that it was neither difficult nor easy for them to enlarge the object, only one participant thinks that it was difficult and only one participant thinks that it was very difficult to solve this task. Most of these participants (with ratings 1–3) are without disabilities and belong to the age group of 55 years and older.

For the *Rotation* task, most participants answered that it was very easy (72.22%) or easy (14.81%) to rotate the virtual object with the assigned interaction (Figure 15). A smaller proportion of the participants think that it was neither difficult nor easy for them to do the rotation (4.63%), while 6.48% of the participants think that it was difficult for them and 1.85% of the participants think it was very difficult to solve this task. Among the respondents who gave ratings 3 or lower, there is an equal share of participants with

disabilities and without disabilities. In addition, among those participants with disabilities, there are mostly people with cognitive and neurological disorders and/or specific learning difficulties, as well as people who, in addition to the above, have physical difficulties or physical disabilities and visual impairments. The respondents without disabilities are participants from different age groups.

For the analysis of the subjective evaluation of interaction mechanisms in transformation tasks in SG2, descriptive statistics were calculated ($N = 108$). The calculated measures (average value M , median C , standard deviation Sd , minimum and maximum value, and confidence interval $CI = 95\%$) are shown in Table 11. Figure 16 shows a graph with mean values of ratings of each transformation task with a confidence interval of 95%. These results provide insight into participants' subjective evaluation of the interaction mechanisms, complementing the descriptive statistics in Table 11.

Table 11. Descriptive statistics for the subjective evaluation of interaction mechanisms in transformation tasks in SG2 ($N = 108$).

	Translation	Scaling	Rotation
M	4.34	4.73	4.49
C	5	5	5
Sd	0.9	0.65	0.98
min	1	1	1
max	5	5	5
CI = 95%	4.17; 4.51	4.61; 4.86	4.3; 4.68

Note: M —mean, C —median, Sd —standard deviation, min —minimum, max —maximum, $CI = 95\%$ —95% confidence interval.

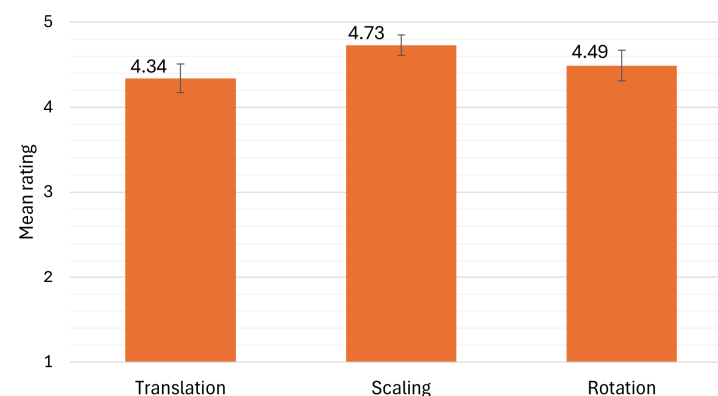


Figure 16. Mean ratings of interaction mechanisms in transformation tasks in SG2 with 95% confidence intervals.

The results indicate that the vast majority of participants found the interaction mechanisms assigned to them for the virtual object transformation tasks easy or very easy, which can also be seen from the mean ratings per task. Since the goal of universal design is to enable the use of the solution by as many users as possible, although it is a smaller share of participants, participants who gave a rating of 3 or lower were additionally analysed to identify problems that prevent the user from performing a particular task.

5.5. Comparison of Subjective Ratings of Interactions in SG1 and SG2

To assess whether the method for determining the interaction mechanism settings is appropriate for the participants, we compared their ratings of interaction mechanisms from SG1 and SG2 ($N = 107$). A participant who did not rate at least one of the interaction

mechanisms was excluded from the analysis. Assumptions behind the calculation of interaction mechanism for a particular task are that if a user skipped the task, then it is considered difficult for them, as well as that they gave the lower rating for such interaction mechanism. Also, it is considered that if a user rated one interaction mechanism of the related tasks with a higher rating, they prefer it more so that interaction mechanism is then assigned to them. In the end, if the ratings are the same for both interaction mechanisms of the related task, then it is considered that the more difficult interaction took them more time (and based on that they were assigned an alternative interaction in SG2). Based on these assumptions, the participant in SG2 should be more comfortable with the assigned interaction in a particular transformation task (compared to an alternative interaction), and the participant would give it a higher (or equal) rating compared to the average of the ratings given for interactions in equivalent tasks in SG1. The equation that describes the above written is:

$$I_2 \geq I_1, I_1 = \frac{\overline{I_{TG}} + \overline{I_{IC}}}{2} \quad (1)$$

$\overline{I_{TG}}$ and $\overline{I_{IC}}$ are the mean ratings of the participants for different interaction mechanisms (TG = Touch Gestures and IC = Indirect Controls) for the same transformation task in SG1, I_1 are the mean values of the ratings I_{TG} and I_{IC} for one transformation task, and I_2 the mean score of interactions for the equivalent transformation task in SG2. The transformation task in SG1 refers to the tasks of translation (related tasks Translation TG-Translation IC), scaling (related tasks Scaling TG-Scaling IC), and rotation (related tasks Rotation TG-Rotation IC).

By subtracting the mean values of the ratings I_1 from I_2 ($I_2 - I_1$) we can compare the participants' ratings of interaction mechanisms in transformation tasks from SG1 and SG2. Figure 17 shows a chart with the results which shows a positive value for the Rotation and Scaling tasks. This indicates that the participants were, on average, more satisfied with the interaction mechanisms in these tasks in SG2 than with the interaction mechanisms in the equivalent tasks in SG1. A negative value for the translation task indicates that the participants were on average more satisfied with the interaction mechanisms in SG1 compared to the assigned interaction mechanism in SG2. To determine whether the obtained difference is statistically significant, the Wilcoxon signed-rank test was performed ($N = 107$). This test is appropriate since participants rated the interaction mechanisms for the same transformation tasks in both SG1 and SG2, creating paired ratings, which is suitable for this non-parametric test which does not require the assumption that the data follow a normal distribution. First, for each user, the average of the ratings assigned to interactions for pairs of tasks (Translation 1, Scaling 1, Rotation 1) in SG1 was calculated and these were compared with the ratings of interactions in the same tasks in SG2 (Translation 2, Scaling 2, Rotation 2). The results of the test are shown in Table 12.

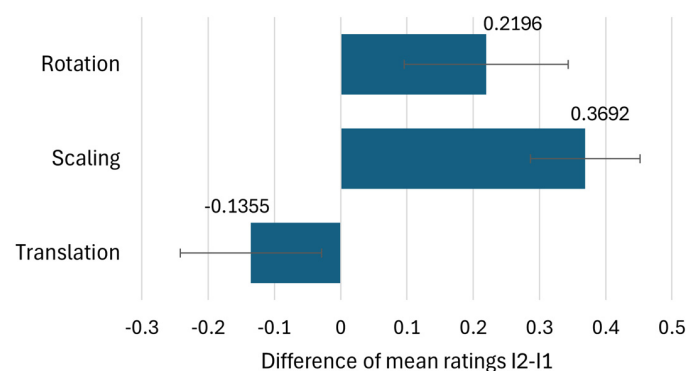


Figure 17. The difference in mean values of interactions' ratings $I_2 - I_1$ for each task.

Table 12. Wilcoxon signed-rank test results for comparison of interaction ratings in SG1 and SG2.

	W	z	p	r
Translation 2—Translation 1	978	−1.209	0.227	-
Scaling 2—Scaling 1	390	−4.55	<0.001	−0.44
Rotation 2—Rotation 1	913.5	−2.575	0.010	−0.25

As can be seen from the table, the difference in the subjective ratings of the interactions for the Translation task is not statistically significant ($W = 978$; $p = 0.227$). Therefore, no conclusions can be drawn regarding the method of assigning the interaction mechanism to an individual participant for the translation task. On the other hand, the difference in the subjective ratings of the interactions for the Scaling ($W = 390$; $p < 0.001$; $r = -0.44$) and Rotation ($W = 913.5$; $p = 0.010$; $r = -0.25$) tasks is statistically significant. Therefore, it can be concluded that the participants were generally more satisfied with the interaction mechanisms in the rotation and scaling tasks in SG2 than in SG1, i.e., the method of assigning the interaction mechanism for these tasks is appropriate.

5.6. Discussion and Limitations

Achieving specified goals effectively, efficiently and with satisfaction impacts not only the usability of a product but also its accessibility, especially when considering users with disabilities. While *usable accessibility*—an approach ensuring that a product is both technically and functionally accessible—is commonly discussed in the context of web design [77], we believe that it can be applied to serious games based on handheld AR. Similar to how the Web Content Accessibility Guidelines (WCAG) 2.0 [78] can be applied to non-web information and communication technologies [79], these guidelines can direct accessible interaction design in AR environments.

In particular, our study aligns with the *Operable* WCAG principle, which emphasizes that user interfaces should be easy to navigate and interact with for a wide range of users, including those with disabilities [78]. One of the WCAG guidelines related to the *Operable* principle states that users should be able to operate functionalities through multiple input methods beyond just keyboard input [80]. For example, complex touch gestures, such as drag-and-drop and swiping, should have alternative methods of interaction to accommodate users who may struggle with these gestures. Applying this guideline to transformation tasks in handheld AR, we introduced indirect controls, i.e., buttons, as an alternative to touch gestures. This provides an accessible option for users who cannot perform certain gestures, either permanently (e.g., due to motor impairments) or situationally (e.g., due to different situational circumstances).

Our user testing results indicate that most participants were satisfied with their assigned interaction mechanisms for interacting with the virtual object. Notably, most of the participants were assigned indirect controls, which proved beneficial for transformation tasks that were more challenging (i.e., rotation and scaling), as reflected in the results for interaction parameters such as completion time, skipped tasks and interaction satisfaction ratings (Section 5.2). These findings highlight the importance of offering multiple interaction mechanisms to accommodate diverse user needs.

However, preferences for interaction mechanisms vary. Some users may find touch gestures more intuitive, while others prefer indirect controls depending on the context and task. Thus, it is also crucial to ensure that indirect controls do not interfere with AR content and can be removed when unnecessary. The ideal interaction experience should be adaptable. For example, users with motor impairments may require larger indirect controls

to prevent unintended touches, while users with visual impairments may find on-screen buttons distracting or obstructive on small mobile screens.

To enhance accessibility and user experience, AR systems should dynamically adjust interaction mechanisms based on user needs and context. By tracking relevant interaction parameters alongside contextual data, systems can recognize when specific mechanisms are needed (or unnecessary) and adjust accordingly. For example, a system could automatically remove visual buttons for users relying on audio cues or enlarge indirect controls for users with motor impairments. However, the best approach to adapt the solution to the user needs to be investigated more in terms of continuous adaptations while considering the risk of user dissatisfaction caused by sudden adjustments.

In summary, findings from this study verified that offering multiple input methods (e.g., touch gestures and indirect controls) and ensuring adaptability based on user interaction data and preferences can be beneficial in the context of *accessible usability*. Game developers should take these recommendations into consideration when developing games based on AR to contribute to digital inclusion and widen their audience. This can also benefit educators who use AR games as educational tools in addition to traditional learning methods. With diverse student needs accommodated, equitable access to educational content can be ensured. However, accessibility in AR-based serious games goes beyond providing alternative input methods. It requires a deeper understanding of how usability, user experience, and accessibility intersect. Designing flexible, user-centred interaction mechanisms ensures not only functional usability but also an enjoyable and satisfying experience for a diverse range of users.

Having said that, one of the limitations of this study is related to recruitment of participants with diverse needs. Sample sizes of some groups of participants with disabilities and of different ages were not sufficient to draw conclusions for the entire population. This primarily refers to the sample of participants in the category of visual impairments, which partly results from technological limitations as the game engine in which serious games were developed (Unity) does not support text-to-speech in the Croatian language, and therefore people who are blind or have low vision and use a screen reader could not participate in the testing. While our study included 108 participants, a larger and more diverse sample could provide further insights into how individual differences (primarily age and disabilities) in combination with data about user interaction can influence conclusions about accessibility and appropriate adaptations in AR-based serious games. In addition to sample size, a study was conducted using two models of tablet devices which may be considered as another study limitation since they do not fully represent the range of devices used in real-world scenarios. Differences in screen size, processing power, and tracking accuracy could affect interaction performance.

6. Conclusions and Future Work

Although solutions based on AR have already become part of our everyday lives, not enough attention was paid to the accessibility of such solutions to make them available to as many users as possible. This is even more important considering the benefits of such solutions in different domains, especially in education where different serious games based on AR can be useful to many learners since they bring innovative and interesting ways of learning. To make AR-based serious games more inclusive, it is important to understand the needs of different users in AR in the context of accessibility. One approach to better understanding the needs of various users in AR is to utilize its interactive nature making it a rich source of data. Tracking user interactions in AR-based serious games can benefit all stakeholders involved in the process of creating an accessible AR for all. By analysing stored data on user activities and results, one can better understand user behaviour and user

performance in AR and thus use this data to adapt the content and interaction mechanisms to the individual user. To be able to make these solutions accessible to as many people as possible, it is inevitable to include people with various disabilities and different ages in the process of designing alternative ways of interacting and perceiving the AR content.

By following universal design principles, it is possible to address many accessibility barriers. However, since there is a need to address the various needs of different users, including people with disabilities and older people which are a very heterogeneous group of users, some requirements to address these needs can be conflicting. To enable flexibility in use and adaptation to an individual user, we proposed a solution that follows the previously mentioned approach of tracking user interactions. We proposed a framework for tracking user interactions to improve the accessibility of serious games based on handheld AR. The framework relies on the proposed methodology for automated collection, storage and analysis of interaction parameters in the context of accessibility. To enable the goal of improving the accessibility of AR-based serious games for an individual user, we first investigated the factors that can have an impact on accessibility and defined a data model for tracking user interaction data based on features of accessibility, serious games and AR for handheld devices. By implementing and evaluating the technological platform based on the proposed methodology and model, we showed that by tracking and analysing user interaction data, it is possible to define adaptation mechanisms that can improve accessibility and user satisfaction when using AR-based serious games. In this paper, special emphasis is placed on the evaluation of interaction mechanisms in transformation tasks in AR that were adapted for the participants in the experiment. The results showed that for the majority of participants, the assigned interaction mechanisms for transformation tasks were satisfactory.

Currently, the method for calculating the interaction settings for an individual user considers selected interaction parameters (objective and subjective metrics). However, the defined data model enables tracking numerous other interaction parameters and the interaction tracking model can be easily expanded with new parameters. Therefore, the system implemented in this way is suitable for further research in the context of determining the impact of other parameters on accessibility and usability as well as defining new guidelines for the development and design of AR-based serious games. Considering the complexity of the research area in the context of the number and combination of different variables that can affect the accessibility and user experience of serious games in AR on mobile devices, for future work we propose to use more advanced inference techniques based on data such as different machine learning methods for determining how to adapt these solutions to individual users. Although it is important to have the possibility that users can independently adjust the solution according to their preferences at any time, sometimes users cannot do this themselves due to various limitations. Therefore, the automatic adaptation of the solution to an individual user proves to be a powerful tool in achieving accessibility, especially for users with disabilities who do not tend to adapt settings manually. The novel uses of technologies such as artificial intelligence open the possibility for future AR serious games to be more accessible and engaging by generating recommendations and automatically adapting them to individual needs.

In addition, the framework proposed in this paper can serve as a foundation for improving accessibility not just in serious games based on handheld AR, but also across various immersive technologies, such as virtual and mixed reality. In this case, the model for tracking user interactions should be revised so that the features and interaction parameters correspond to tasks and interaction techniques usual for these technologies as well as the context in which the solution is being used (related to the virtual environment or the device itself).

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Appendix A

Table A1 brings defined key points of interaction in serious games based on augmented reality (AR) with defined triggers that initiate events, i.e., user interaction tracking. Defining key points of interaction is a first step when specifying a model for tracking user interactions.

Table A1. Key points of interaction in AR-based serious games with defined triggers that initiate user interaction tracking.

Key Interaction Point	Event Trigger
Launching a serious game	The user launched a serious game on the device
Instructions page opened	The user pressed a button to open the instructions page
Instruction page read	The user scrolled to the end of the instruction page
Game started	The user pressed the “Game” button
User login	The user pressed the continue button after entering the correct username
Starting an AR session in a game	The user pressed a button to start an AR session
Starting AR scene setup	The user has started scanning by clicking a button
Physical/virtual correspondence established	The user has established an AR scene
Game level started	The user pressed the button to start the game level
Virtual object selected	The user selected a virtual object for interaction
Interaction with a virtual object	The user interacted with a virtual object
Game level completed	The user completed the game level
Game level skipped	The user skipped the level by pressing the skip button
Game session exited	The user confirmed the exit from the current game session
Game session completed	The user completed all levels in a game session

Table A1. Cont.

Key Interaction Point	Event Trigger
Settings selected	The user entered settings by pressing on the settings button
Settings changed	The user pressed the button to save the settings
Freeze option turned on	The user has turned on the freeze option by pressing the freeze button
Help requested	The user opened additional instructions by selecting the help button
Interaction rated	The user has selected a rating for the interaction

The next step is the specification of interaction parameters which is based on an overview of common solutions in AR considering the features related to the user, serious games, augmented reality, context and accessibility (see more details in Section 3.1). Table A2 shows specified interaction parameters with their descriptions and data types. The values of the parameters depend on the data type defined for the parameter. One specific interaction parameter is related to the interaction type and the number of times the interaction was detected, i.e., pair *interaction-type:number* (*string:int*). Interaction type corresponds to the input method usual for AR, i.e., touch gestures and indirect controls (*drag*, *button_up*, *button_down*, *button_left*, *button_right*, *button_rotate_left*, *swipe_left*, *button_rotate_right*, *swipe_right*, *button_increase*, *zoom*, *button_decrease*, *pinch*, *two_finger_rotate*).

Table A2. Interaction parameters specified for tracking events in the context of accessibility.

Interaction Parameter	Data Type	Description
<i>duration</i>	<i>float</i>	Time spent in a certain game level (s)
<i>times-played</i>	<i>int</i>	Number of game plays
<i>score</i>	<i>float</i>	The ratio of correct and incorrect answers/the final score
<i>used-help</i>	<i>int</i>	Number of help requests
<i>skipped</i>	<i>boolean</i>	Flag indicating whether the task was skipped
<i>screen-size</i>	<i>string</i>	Device screen size (width x height)
<i>ambient-intensity</i>	<i>float</i>	The value of the light intensity in the environment
<i>ambient-noise</i>	<i>float</i>	The value of ambient noise in the environment
<i>target-distance</i>	<i>float</i>	Distance between the user (device) and the target
<i>condition-value</i>	<i>float/int</i>	The value indicating the condition to be achieved, e.g., time (s) or size
<i>object-distance</i>	<i>float</i>	Distance between the positions of two virtual objects
<i>time-for-interaction</i>	<i>float</i>	Time required for the user to interact (s)
<i>progress</i>	<i>float</i>	Percentage of progress (value between 0 and 1), e.g., progress in the game or interacting with the object
<i>interaction-type, number</i>	<i>string:int</i>	The type of interaction that was used and the number of times the interaction was detected
<i>rotate-direction</i>	<i>string</i>	The direction in which the user rotates the virtual object (<i>left</i> , <i>right</i> , <i>none</i>)
<i>attempt-number</i>	<i>int</i>	Number of attempts

Table A2. Cont.

Interaction Parameter	Data Type	Description
<i>change-in-distance</i>	<i>float</i>	The ratio of the difference in the change in the distance of the device's camera with the initial distance from the virtual object and the initial distance from the virtual object (the device can move away or get closer to the virtual object)
<i>out-of-fov-time</i>	<i>float</i>	The time that the virtual object being observed is not in the field of view (s)
<i>scaled-item</i>	<i>float</i>	Virtual object size (in game-specific units)
<i>freeze</i>	<i>boolean</i>	Flag indicating whether the <i>freeze</i> option was used in a certain task or game level
<i>freeze-enabled</i>	<i>boolean</i>	Flag indicating whether the <i>freeze</i> option is enabled
<i>scale-text</i>	<i>boolean</i>	Flag indicating if <i>scale text</i> option is used
<i>text-size</i>	<i>float</i>	Text size (increment steps depend on the game)
<i>change-font</i>	<i>boolean</i>	Flag indicating whether the <i>change font option</i> is used
<i>used-font</i>	<i>string</i>	Text font at the end of the game
<i>contrast-combination</i>	<i>int</i>	Colour contrast between text and the background
<i>audio-support</i>	<i>boolean</i>	Flag indicating whether audio support is used
<i>button-size</i>	<i>float</i>	Button size (increment steps depend on the game)
<i>high-contrast</i>	<i>boolean</i>	Flag indicating whether high contrast of the virtual object is used
<i>edge-highlight</i>	<i>boolean</i>	Flag indicating whether edge-highlighting of the virtual object is used

After the specification of interaction parameters, it is necessary to define the data structure of previously specified events, i.e., key interaction points. Since the model for tracking user interactions in AR-based serious games proposed in this work is based on the interaction model defined in [58] (see more in Section 3.1.3), it is necessary to map the event attributes and previously defined interaction parameters to xAPI statements. To recall, the interaction event is defined with the following attributes: timestamp, user identifier, action, target, and optional value. For example, if we are tracking the event when the user launched the game, the event trigger for this interaction point is *the user launched a serious game on the device*. In this example, the *action* attribute corresponds to the verb *launched*, while the *target* corresponds to the object type presenting a *serious game*. Possible *optional values* in this event could be various contextual information, such as information about the device and/or time of the day. The interaction event attributes of a serious game are mapped to xAPI statement attributes as shown in Table A3.

Table A3. Mapping of interaction event attributes to xAPI statement attributes (adapted from [58]).

Interaction Event Attribute	xAPI Statement Attribute
user id	actor
action	verb
target	object
optional value	result, context
timestamp	timestamp

The xAPI standard identifies verb and object attributes with a unique Internationalized Resource Identifier (IRI), as well as the extensions of statements within the result and context attributes. In addition, we bring possibilities of how different interaction event attributes can be mapped to follow the xAPI standard and to cover different aspects of AR-based serious games in the context of accessibility. Table A4 shows the mapping of xAPI verbs (represented by IRIs) which can be used to cover actions of interaction events with an example of a corresponding key interaction point in an AR-based serious game earlier specified. Description of each xAPI attribute (e.g., verb) is available on the xAPI Profile Server (<https://profiles.adlnet.gov/> (accessed 15 December 2024)).

Table A4. Mapping of xAPI verbs represented by Internationalized Resource Identifiers to selected key interaction points.

xAPI Verb	Key Interaction Point
http://adlnet.gov/expapi/verbs/launched	Launching a serious game
https://w3id.org/xapi/seriousgames/verbs/pressed	Instructions page opened
https://w3id.org/xapi/adb/verbs/read	Instruction page read
https://w3id.org/xapi/dod-isd/verbs/input	User login
http://adlnet.gov/expapi/verbs/initialized	Starting an AR session in-game
https://w3id.org/xapi/seriousgames/verbs/accessed	Starting AR scene setup
https://w3id.org/xapi/dod-isd/verbs/scanned	Physical/virtual correspondence established
http://id.tincanapi.com/verb/selected	Virtual object selected
https://w3id.org/xapi/dod-isd/verbs/performed	Interaction with a virtual object
http://adlnet.gov/expapi/verbs/preferred	Settings changed
http://adlnet.gov/expapi/verbs/completed	Game level completed
http://id.tincanapi.com/verb/skipped	Game level skipped
http://adlnet.gov/expapi/verbs/exited	Game session exited
http://adlnet.gov/expapi/verbs/terminated	Game session completed
https://w3id.org/xapi/dod-isd/verbs/assessed	Interaction rated

Table A5 shows the most common targets that can appear in AR-based serious games on a mobile device. Examples of possible objects or activities in a serious game are mapped to corresponding xAPI objects.

The event attribute *optional value* can be mapped to the xAPI *result* and *context* properties. This mapping allows us to track interaction parameters specified earlier. Because some of the parameters are specific to the accessibility domain, the corresponding xAPI properties related to *result* or *context* attributes do not exist and must be defined. However, some of them are sufficient and can be used. For example, the *context* attribute contains a *platform* property that indicates the type of platform used for the experience, and the *result* attribute contains the *score*, *success*, *completion*, *response* and *duration* properties, which can be used to track game progress, but also success or more detailed insights into user interaction, e.g., how long the user spent on a certain interaction, the number of attempts and which interactions were used. For the remaining interaction parameters, it is necessary to define new *extensions*, if they are not defined in existing xAPI profiles. Table A6 lists examples of *extensions* related to *result* and *context* that can be mapped to some of the previously defined interaction parameters important to track in the context of accessibility.

Table A5. Mapping of xAPI objects represented by Internationalized Resource Identifiers to selected objects/activities in a serious game.

xAPI Object	Object/Activity
https://w3id.org/xapi/seriousgames/activity-types/menu	Option in the main menu
https://w3id.org/xapi/seriousgames/activity-types/item	A virtual object in an AR scene
http://adlnet.gov/expapi/activities/cmi.interaction	Username input
https://w3id.org/xapi/seriousgames/activity-types/level	Game level or task
https://w3id.org/xapi/seriousgames/activity-types/serious-game	Serious game
https://w3id.org/xapi/acrossx/activities/page	Instructions page
http://adlnet.gov/expapi/activities/attempt	New game/AR session attempt
http://id.tincanapi.com/activitytype/scenario	AR scene

Table A6. Mapping of xAPI result and context extensions represented by Internationalized Resource Identifiers to selected interaction parameters.

IRI of Result/Context Extensions	Interaction Parameter
https://w3id.org/xapi/video/extensions/screen-size	<i>screen-size</i>
http://new-xAPI-profile-domain/result-extension/used-help	<i>used-help</i>
http://new-xAPI-profile-domain/result-extension/skipped	<i>skipped</i>
http://new-xAPI-profile-domain/context-extension/environment	<i>ambient-intensity</i>
https://w3id.org/xapi/acrossx/extensions/starting-point	<i>change-in-distance</i>
http://id.tincanapi.com/extension/ending-point	
http://new-xAPI-profile-domain/result-extension/rotate-direction	<i>rotate-direction</i>
http://new-xAPI-profile-domain/result-extension/out-of-fov-time	<i>out-of-fov-time</i>
http://new-xAPI-profile-domain/result-extension/object-distance	<i>object-distance</i>
https://w3id.org/xapi/seriousgames/extensions/progress	<i>progress</i>

Appendix B

Table A7 details the questionnaire used for the subjective evaluation of the serious game (SG2) based on augmented reality. The questions were compiled considering the principles of Web Content Accessibility Guidelines 2.0 [78], ease of use related to certain interactions in the game, satisfaction with the way of interacting in the game, and the need for additional accessibility options. The list of questions was supplemented with questions related to the user experience of a serious game with questions selected from the User Experience Questionnaire (UEQ) (<https://www.ueq-online.org/> (accessed 15 December 2024)). It should be noted that all questions in the questionnaire were designed to be understandable by people with disabilities, especially people with intellectual disabilities. Therefore, simple terms were used in the questions, and the questions and answers taken from standardized questionnaires were adapted accordingly. Questions and answers relevant to this paper's results are bolded.

Table A7. Questions for subjective evaluation by sections with answer options.

Question	Options
Please specify your gender.	Female; Male; Prefer not to say
Which age group do you belong to?	Under 18; 18–24; 25–34; 35–44; 45–54; 55–64; 65+
Please select the statements that apply to you.	I am a person with visual impairments.; I am a person with hearing impairments.; I am a person with motor disabilities.; I am a person with cognitive and neurological disorders.; I am a person with specific learning difficulties.; None of the statements apply to me.
If you have visual impairments/hearing impairments/motor disabilities/cognitive and neurological disorders/specific learning difficulties, please describe them.	(Free-form answer)
Do you use a smartphone or tablet in your daily life?	Yes; No
How often do you use your smartphone or tablet?	Every day; 5 days a week; 2–3 days in a week; Sometimes
Have you used applications with augmented reality before?	Yes; No
How often have you used augmented reality applications before?	Once; A few times; Often; Every day
Is the font used in the game easy to read?	
Is the font used in the game big enough to read?	
Is the text easy to read considering the colour of the text and background?	Yes; No
Have you used the options to change the font type and size in the settings?	
How difficult or easy was it for you to change the font type and size in the settings?	1 = Very difficult; 2 = Difficult; 3 = Neither difficult nor easy; 4 = Easy; 5 = Very easy
Are the virtual objects that appear in the game noticeable enough concerning the background, i.e., the surface of the room in which you are using the solution?	Yes; No
How difficult or easy was it for you to scan the surface before a car was created on it?	1 = Very difficult; 2 = Difficult; 3 = Neither difficult nor easy; 4 = Easy; 5 = Very easy
What would make scanning the surface in the game easier for you?	Additional textual instructions on how to scan would be easier.; An animation showing how to scan would make it easier.; Nothing, it is easy to select; Other
How difficult or easy was it for you to change the colour of the car?	
How difficult or easy was it for you to move the car to the desired location?	1 = Very difficult; 2 = Difficult; 3 = Neither difficult nor easy; 4 = Easy; 5 = Very easy
How difficult or easy was it for you to scale the car?	
How difficult or easy was it for you to rotate the car?	
How difficult or easy was it for you to enter the car registration numbers in the last task?	
Did you use buttons (to move, rotate, or zoom) in any car task?	Yes; No
Are the buttons for controlling the car big enough?	
Are the buttons for controlling the car in the appropriate place?	
Did you use the Skip button in any task?	
Select the reason for skipping the task:	The task was difficult.; The task was not interesting.; I wanted to try out what would happen when I click the Skip button.; Other
Are the instructions for each task clear and understandable (is the user well informed about what to do in each task)?	Yes; No

Table A7. Cont.

Question	Options
Are you satisfied with the way you controlled the car?	1 = Very dissatisfied; 2 = Somewhat dissatisfied; 3 = Neither satisfied nor dissatisfied; 4 = Somewhat satisfied; 5 = Very satisfied
The following 4 questions contain pairs of opposite characteristics that may apply to the serious game (SG2). Please select the degree of agreement with the characteristic of the game that most closely reflects your impression.	Frustrating 1 2 3 4 5 Fun Incomprehensible 1 2 3 4 5 Understandable Complicated 1 2 3 4 5 Simple Unattractive 1 2 3 4 5 Attractive
In case of any additional comments and suggestions, please state them.	(Free-form answer)

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